

A Shared Template Without Shared Feedback: Funding Rates in Cryptocurrency Perpetual Futures

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Abstract

Perpetual futures lack maturity, so exchanges use funding to keep prices close to spot. We develop a model with extrapolative traders, capital-constrained arbitrageurs, liquidity suppliers, and a piecewise funding rule. The model yields a capital-dependent stability band. Weak feedback leaves price gaps persistent, intermediate feedback stabilizes them with or without crossing spot, and feedback that is too strong generates expanding reversals. Losses and liquidation reduce future corrective capacity and can make stabilization infeasible. A binding leverage constraint can also prevent stronger funding from expanding corrective positions. Pre-settlement trading, interval averaging, and the rule's flat and capped regions separate the nominal template from the funding response delivered at settlement. Evidence from 200 Binance USDT-margined perpetuals supports the model's implications that the nominal funding rule does not by itself determine effective feedback, basis reversals are associated with directional liquidation, and rate caps accompany persistent price gaps. These results help explain why a shared funding template does not guarantee shared feedback.

Keywords: perpetual futures; funding rate; futures basis; limits to arbitrage; market design.

JEL classification: G12, G13, G14, D47.

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1 Introduction

Since the first Bitcoin perpetual futures contract appeared in 2016, perpetuals have become the dominant derivative instrument in cryptocurrency markets, surpassing dated futures in trading volume on major venues such as Binance, OKX, and Bybit. They have no maturity date that mechanically pulls their price toward spot. Exchanges instead rely on scheduled funding transfers between long and short positions.¹ When the contract trades rich, longs typically pay shorts, whereas shorts typically pay longs when it trades cheap. Funding is designed as a negative-feedback mechanism. Yet its equilibrium effect need not stabilize the basis. This paper asks when funding closes the price gap, when it produces a stable reversal in which the perpetual crosses spot while the gap continues to shrink, and when it undercorrects or generates expanding reversals.

The cross-section shows why this question matters. Binance uses a shared funding template for many contracts. Contract-specific caps, settlement intervals, maintenance-margin parameters, and some interest-rate settings vary across contracts and dates. Yet reset-to-reset basis dynamics differ far more than this parameter variation suggests. We sort the 200 USDT-margined perpetuals in our sample into capacity-proxy quintiles using each contract’s median open interest. From the deepest quintile to the thinnest quintile, the average absolute basis rises from 6.5 to 9.1 basis points and economically meaningful basis reversals rise from 1.2% to 4.3% of funding resets. These contrasts motivate the central puzzle: how can a shared funding template coexist with such different market dynamics?

Our explanation is that market trading transforms the exchange’s posted formula into effective feedback. By effective feedback, we mean the reduced-form slope from the basis at the start of an interval to the funding rate settled at its end. Extrapolative speculative demand can keep a price gap alive (De Long et al., 1990; Barberis et al., 2018). Capital-constrained arbitrageurs trade against the gap (Shleifer and Vishny, 1997; Gromb and Vayanos, 2002). Liquidity suppliers turn the remaining order imbalance into price changes (Grossman and Miller, 1988; Hendershott and Menkveld, 2014). Effective feedback therefore depends on market depth, arbitrage capital, whether the rule is on its flat, active, or capped branch, and settlement timing. The same funding template can produce

¹On Binance USDT-margined perpetual contracts, the rate combines an interval-average premium index with an interest component through a clamp and is subject to contract-specific cap and floor limits. See Binance’s [funding-rate rules](#) and its developer documentation for [contract-specific caps, floors, and settlement intervals](#).

different effective feedback across contracts and states.

To examine this mechanism in detail, we build an equilibrium model with two linked time scales. Trading and price formation occur continuously within each funding interval. Funding settles and arbitrage capital updates at scheduled boundaries. Speculative traders reinforce the perpetual–spot price gap. Capital-constrained arbitrageurs take offsetting positions, and liquidity suppliers absorb the remaining pressure. At settlement, the exchange applies its clamp-and-cap rule to a same-interval average of the premium-index input. Trading profits and liquidation following a margin shortfall determine capital for the next interval. Capital in turn determines how much corrective trading arbitrageurs can supply. News and other order-flow shocks also move the basis. We call the part of the opening dislocation carried into the next reset the inherited dislocation. For the stability analysis, a model gain summarizes the local market response to funding.

The central stability result is that funding can be too weak or too strong. In a stressed market, the price gap would grow without funding. When the model gain is too low, the inherited dislocation grows. At an intermediate model gain, it shrinks. When the model gain is too high, it reverses sign and grows in absolute value. The stability band is the range of model gains that makes the inherited dislocation shrink. Within that band, convergence may remain on the same side of spot or may alternate across it. The flip threshold marks the change between these two patterns. This is why a sign change alone does not imply instability. It is also why stronger feedback is not always more stabilizing. In stressed states, stability requires a positive minimum model gain. That minimum is zero when untreated market forces do not expand the basis.

Arbitrage capital determines where the stability band lies. When the arbitrageurs' leverage constraint is slack, more capital lets them absorb more speculative pressure. It therefore lowers the minimum model gain required in stressed states. Losses and liquidation work in the opposite direction. If this minimum reaches or exceeds the maximum feasible model gain, no feasible gain can stabilize small deviations on the model's interior branch. We call this outcome local interior-branch infeasibility. The intuition is simple. The market cannot produce enough correction. A binding leverage constraint can provide a separate route to the same problem. Once the constraint binds, stronger funding need not produce a larger offsetting position. Fragility is therefore intertemporal. A loss today can make the same model gain too weak tomorrow. In a benchmark version of the model, liquidation creates a ratchet. Liquidation lowers capital and raises the minimum stabilizing

model gain, so a large liquidation episode can move the market across the infeasibility boundary.

The nominal template derivative is not the same as effective feedback. Traders act before settlement. The exchange therefore applies its formula to the interval-average premium, not to the opening basis. The flat branch and the rate cap further change this relation. We call the full link from the opening basis to the settled funding rate the state-to-settlement mapping. For measurement, we separate this mapping into two observable parts. Template pass-through measures how strongly the formula converts the interval-average premium into the settled funding rate. Interval retention measures how much of the opening basis remains in that average. Put simply, one part measures the rule, and the other measures how much of the opening basis reaches the rule’s input. Together, they explain why the same nominal template derivative can produce strong effective feedback in one contract or state and weak effective feedback in another.

The model also gives results on liquidation and the funding interval. It predicts that a funding-induced reversal can shift margin pressure toward the side opposite the opening basis. If the perpetual begins above spot and later crosses below it, for example, leveraged longs become more exposed to liquidation. Liquidation can then reduce the capital available for later corrective trading and make the market more fragile. When the model gain per interval is held fixed, a shorter funding interval speeds convergence if the basis stays on the same side of spot. It can instead increase back-and-forth adjustment when the basis alternates across spot. The simulations illustrate the stability band, the liquidation ratchet, and the resulting capital dynamics. They also examine outside capital entry and show how it helps the market recover from low-capital states. Together, these results show why a model gain that works in one market state may fail in another.

We examine three observable implications of the model using 677,405 contract-by-funding-interval observations for 200 Binance USDT-margined perpetuals from September 2020 through December 2025. We first estimate the state-to-settlement mapping from the opening basis to the funding rate settled at the end of the interval. We then estimate template pass-through using an interval-mean premium-index proxy, which is closer to the input in the exchange’s formula. Although the two estimates use different state variables, both have the same piecewise shape. Each is nearly flat in the central region and positive on either side. In a separate contract-level decomposition using the interval-mean basis proxy, effective feedback closely tracks template pass-through multiplied by interval retention.

Large basis reversals are accompanied by directional margin pressure. Intervals containing an economically meaningful sign change have substantially more liquidation on the side opposite the opening basis. Unusually large absolute funding is associated with more next-interval liquidation on both sides. The association is larger on the side opposite the opening basis. When the funding rate reaches its cap, the formula can no longer adjust the payment further in the direction called for by the premium. Cap hits are about thirteen times more frequent in the thinnest than in the deepest capacity-proxy quintile. They are also associated with greater subsequent basis persistence. This persistence association is concentrated in a small number of contracts with many cap events. Together, these patterns provide evidence on the state-to-settlement mapping, directional margin pressure, and cap-limited feedback.

Our main theoretical contribution is to add a dynamic-stability perspective to the pricing and design of perpetual-futures funding. Existing models characterize no-arbitrage regions, target-tracking rules, and contract design, including [Dai et al. \(2025\)](#), [Kim and Park \(2025\)](#), and [Gornall et al. \(2024\)](#). We complement these results by deriving a capital-dependent stability band for the model gain and an interior flip threshold.

The paper’s second contribution is to show how arbitrage capital affects whether the market can stabilize the basis. The limits-to-arbitrage literature shows how losses and collateral constraints weaken corrective trading ([Shleifer and Vishny, 1997](#); [Gromb and Vayanos, 2002](#); [Brunnermeier and Pedersen, 2009](#)). [Duffie \(2010\)](#) explains why corrective capital can move slowly. [Gornall et al. \(2024\)](#) bring constrained arbitrage capital into the study of perpetual futures. Building on this work, we identify when corrective trading is no longer enough to stabilize the basis.

The final contribution is to make the difference between a posted funding rule and the feedback delivered at settlement measurable. Related work studies clamps and path-dependent funding rules ([Dai et al., 2025](#); [Hugonnier and Jermann, 2025](#); [Kim and Park, 2025](#)). [Gornall et al. \(2024\)](#) study funding frequency. We characterize the state-to-settlement mapping. In the data, we estimate its average slope contract by contract. We also document its piecewise shape and relate that slope to template pass-through and interval retention.

Related literature. A first strand prices perpetual claims and studies the design of their funding rules. [Shiller \(1993\)](#) proposed perpetual futures as long-horizon claims without fixed expiration. [He](#)

et al. (2022) derive no-arbitrage prices and transaction-cost bounds and document sizable, comoving deviations that diminish over time. Akerer et al. (2026) give a random-maturity representation covering linear, inverse, and quanto perpetual contracts, and Angeris et al. (2023) derive model-free funding and discount rates for two classes of perpetual claims. On design, Dai et al. (2025) show that an exchange-style piecewise-linear clamp creates wide model-free no-arbitrage regions that fit Binance data better than fee-only bounds. Hugonnier and Jermann (2025) show that stochastic interest-rate differentials and a clamp-induced inaction region can produce a bimodal futures-to-spot ratio. Kim and Park (2025) use a mechanism-design approach to construct funding-rate functions, including path-dependent versions, that make a perpetual track a target. Gornall et al. (2024) compare perpetual and quarterly futures under constrained arbitrage capital and link funding frequency to basis risk and liquidity. This work clarifies valuation and rule design, but it leaves open how trading before settlement changes the input to the posted rule, whether the feedback delivered at settlement is dynamically stable, and why one posted template can produce different responses across contracts and capital states.

A second strand provides the theories of speculative demand and liquidity supply on which the model's price formation builds. De Long et al. (1990) show how positive-feedback demand can amplify price movements and induce destabilizing rational speculation. Barberis et al. (2018) model demand driven by extrapolation from recent price changes. Grossman and Miller (1988) explain why immediate execution requires compensated inventory holders, and Hendershott and Menkveld (2014) link inventory imbalances to transitory price pressure. Hazelkorn et al. (2023) show that the equity-index futures–cash basis moves with dealer and investor positions and reflects persistent demand for market exposure. Grandmont (1977) provides a temporary-equilibrium framework for sequential trading with expectations formed from current and past states. These theories supply the model's behavioral and market-microstructure foundations, but they do not study how these forces interact with a funding payment calculated from an interval-average premium input.

A third strand studies the dynamics of arbitrage capital. Shleifer and Vishny (1997) show that losses can cause investors to withdraw capital when arbitrage opportunities become most attractive. Gromb and Vayanos (2002) link collateral constraints to arbitrage positions, market liquidity, and prices. Brunnermeier and Pedersen (2009) explain how market liquidity and funding liquidity reinforce each other, and Duffie (2010) shows why slow-moving capital can delay the correction

of price gaps. These studies show why limited capital can prolong or amplify a dislocation, but they do not study how capital carried across funding intervals interacts with a scheduled piecewise funding rule.

A fourth strand provides the cryptocurrency evidence. [Schmeling et al. \(2026\)](#) attribute fixed-maturity crypto-futures carry to trend-chasing demand and limits on arbitrage capital. [Christin et al. \(2022\)](#) document high Sharpe ratios for a long-spot, short-perpetual strategy driven by persistently positive funding and demand for leveraged exposure. [Makarov and Schoar \(2020\)](#) show that transaction costs alone cannot explain persistent cross-exchange price differences and emphasize segmentation and limited capital flows. [Alexander et al. \(2023\)](#) show how leverage and collateral management shape Bitcoin-futures hedging under liquidation risk, and [Kogan et al. \(2024\)](#) document momentum-like cryptocurrency trading by retail investors, opposite to their behavior in stocks and gold. On market quality, [Ruan and Streltsov \(2022\)](#) find that the introduction of perpetual contracts raises spot trading activity while widening spreads, with the effects concentrated around funding settlement times. [De Blasis and Webb \(2022\)](#) document seasonal and opening effects, with cash-and-carry opportunities clustering during dislocations, and [Morin and Moore \(2024\)](#) find close long-run spot–perpetual comovement alongside frequent five-minute deviations. Little of the evidence operates at the contract-reset level, where the mapping from the opening basis to the settled rate, liquidation on the side opposite a reversal, and persistence under a binding cap remain undocumented. Those reset-level links are the empirical bridge to a theory of funding as market feedback.

The paper proceeds as follows. Section 2 presents the institutional rule and the two motivating facts. Section 3 sets out the optimizing environment, solves the within-interval equilibrium, and turns it into reset-to-reset dynamics with the stability band and the flip threshold. Section 4 endogenizes arbitrage capital through trading profits, margin usage, and liquidation and derives the fragility and infeasibility results. Section 5 adds the leverage constraint, the clamp and cap, and the funding interval as design margins. Section 6 constructs the state-to-settlement mapping and its measurable parts, and Section 7 reports the evidence. Section 8 concludes. Proofs, numerical implementation, and extensions appear in the appendices.²

²The print appendix proves every core result. The Online Appendix (“OA”), which circulates with this draft, records the numerical algorithm, complete parameterization, the simulation figures, sensitivity exercises, and additional analytical extensions.

2 Institutional setting and two motivating facts

The funding rule as implemented. On Binance USDT-margined perpetuals, the setting of our data, funding is exchanged every eight hours (some contracts are moved to four- or one-hour intervals). The rate applied to positions at a funding timestamp is computed from the time-average of a premium index over the preceding interval, plus a small interest component, passed through a clamp: within a flat region of the premium (per 8h, roughly $[-0.04\%, +0.06\%]$) the rate equals the interest component regardless of the premium. Outside it the rate tracks the average premium one-for-one. The rate is capped, at $\pm 0.75\%$ under the standard template. Settlement occurs at the end of the interval over which the premium is averaged. Three features matter for everything that follows: the rule is lagged and discrete (premium information is accumulated over an interval and converted into an end-of-interval transfer), it is templated (one hard-coded clamp structure is shared across hundreds of contracts, with contract-specific caps, settlement intervals, and interest settings varying within it), and it is piecewise (the flat region and cap truncate marginal feedback in the smallest and largest states). Cross-sectional outcomes can still differ because the template input, market response, and trader composition differ.

Fact 1: one template, very different outcomes. Outcomes across contracts sharing the template differ far more than the contract-specific parameter variation suggests. The introduction reports two gradients across capacity-proxy quintiles, in the average absolute basis and in the frequency of economically meaningful sign reversals. Two further gradients sharpen the contrast: from the deepest to the thinnest quintile, cap binding rises by a factor of about thirteen, and the rolling reset coefficient falls from 0.132 to 0.089 (Table 3). These are raw sample gradients, and they are what a theory of the rule has to explain.

Fact 2: the flat region is common; cap binding is rare but concentrated. The clamp’s zero-slope regions are not tail events. About a fifth of interval-average premium observations fall inside the interest-centered flat region, where the settled rate equals the interest component regardless of the premium, and cap binding, rare overall at 0.083% of resets, concentrates almost entirely in the thinnest contracts. Any account of how funding acts on the basis therefore has to be piecewise.

Together the two facts fix the theoretical object at the center of the model. Fact 2 says that

object cannot be the headline template: because the rule is flat in one region and capped in another, what acts on the basis is a local feedback slope that maps the reset-time state into the correction actually applied. Fact 1 says that slope cannot be a constant either: outcomes diverge sharply across contracts sharing one template, so the market’s response to a given correction must also depend on a state the exchange does not set. In the model that state is arbitrage capital.

3 Economic environment, equilibrium, and reset dynamics

The model has two time scales. Trading, price formation, and profit realization occur in continuous time within a funding interval. At interval boundaries the exchange settles funding, public information is revealed, and arbitrage capital is updated. The within-interval block derives the propagation and feedback coefficients from optimizing traders, arbitrageurs, and liquidity suppliers. The baseline is a local temporary equilibrium: belief maps are primitive, agents optimize given those maps, and the arbitrage constraint is initially slack. Every objective and equilibrium condition needed for the main mechanism appears here.³

3.1 Assets, timing, and the exchange rule

Calendar time is $t \geq 0$. Funding intervals have common length $\Delta > 0$ and reset dates $\tau_n = n\Delta$, where $n \in \{0, 1, \dots\}$. Table 1 collects the resulting timing convention, which distinguishes what is predetermined at a reset from what is realized during the interval and revealed at its end. There is a spot asset with strictly positive price S_t and a perpetual futures contract with strictly positive price F_t , both in the same currency units. Both log prices load one-for-one on an otherwise unrestricted common valuation process v_t . This component therefore cancels from the dimensionless log basis

$$x_t \equiv \log F_t - \log S_t, \quad X_n \equiv x_{\tau_n}.$$

The reset state X_n is observed immediately after the reset-date public revelation at τ_n , whose information content is specified in Section 3.3. The scalar $z_n \in \mathbb{R}$ is a publicly observed persistent

³Detailed continuum aggregation and signal extraction are in OA App. A. Throughout, a stock such as the basis or capital is measured at a date, a flow such as anticipated carry is measured per unit of calendar time, and a settled funding rate is a dimensionless transfer per unit notional over one interval.

relative-order-flow state, distinct from the common valuation process. Arbitrage capital $W_n > 0$ is the sector's normalized risk-bearing capacity, measured in the same balance-sheet units as profit and margin below. Economically, it summarizes the balance-sheet capacity available to support paired spot–perpetual positions, a risk-bearing stock distinct from book equity. Model arbitrageurs observe it when positions are chosen, and it is latent in the empirical analysis. Both z_n and W_n are predetermined and held fixed on $[\tau_n, \tau_{n+1})$.

The exchange applies its rule to a dimensionless premium-index input $P_t \in \mathbb{R}$, constructed from venue prices and index conventions and distinct by construction from the log basis x_t . The exchange averages this input over the current interval and settles a total per-unit-notional transfer at its end:

$$\bar{P}_n \equiv \frac{1}{\Delta} \int_{\tau_n}^{\tau_{n+1}} P_t dt, \quad f_{n+1}^{\text{set}} = G(\bar{P}_n). \quad (1)$$

Here G is the published piecewise settlement map, with values in $[-\bar{f}^{\text{set}}, \bar{f}^{\text{set}}]$, and $\bar{f}^{\text{set}} > 0$ is the maximum absolute interval transfer. The map is flat around the interest component, locally increasing on either active branch, and capped in the tails. Section 5.2 gives its exact interest-centered clamp and cap when those branches are analyzed. Section 6.1 uses only its locally active, uncapped branch. Thus (1) is an institutional timing definition.

For closed-form theory, the anticipated local carry over interval n is instead benchmarked by the behavioral rule

$$c_n = \frac{\kappa^m}{\Delta} X_n, \quad \kappa^m \geq 0. \quad (2)$$

The scalar c_n is an anticipated carry flow per unit time, known at τ_n and fixed within the interval, with total anticipated transfer $\Delta c_n = \kappa^m X_n$. The gain κ^m is a model object: Section 6 restores the same-interval average on the locally active branch and derives its relation to the exchange template and the empirical effective slope.

3.2 Extrapolative retail demand

All payoff terms in the trader problems are per-unit-time expected returns. Let μ_M be retail traders' extrapolative expected-return schedule, a real-valued function of the basis that is odd, nondecreasing, continuously differentiable, with $\mu_M(0) = 0$, $\mu'_M(0) = \beta_M > 0$, and finite saturation $\sup_x |\mu_M(x)| =$

$\bar{\mu} < \infty$. A continuum of atomistic traders $i \in [0, 1]$ observes noisy signals $s_{i,t} = \mu_M(x_t) + \gamma z_n + \xi_{i,t}$, with $\xi_{i,t}$ idiosyncratic interpretation noise and $\gamma \in \mathbb{R}$ mapping the forcing state into relative perpetual demand, and chooses signed perpetual notional under quadratic capacity cost,

$$\max_{q \in \mathbb{R}} \left\{ q(s_{i,t} - c_n) - \frac{q^2}{2\eta_i} \right\}, \quad \eta_i > 0. \quad (3)$$

Under a capacity-weighted cancellation condition, aggregation is exact and local linearization gives

$$q_t^M = \eta_M [\mu_M(x_t) + \gamma z_n - c_n] = \eta_M (\beta_M x_t + \gamma z_n - c_n) + o(x_t),$$

with $\eta_M \equiv \int_0^1 \eta_i di > 0$ aggregate retail capacity.⁴ Positive β_M is the extrapolative force: a rich perpetual attracts additional perpetual demand. Funding enters with the opposite sign because carrying that position is costly. The saturating global map prevents the local linear benchmark from mechanically generating unbounded demand in tail states.

3.3 Public forcing and news

The public forcing state follows the reset-time law

$$z_{n+1} = \phi_z z_n + \vartheta_{n+1}, \quad \phi_z \in (-1, 1). \quad (4)$$

The scalar innovation ϑ_{n+1} is not in the public reset-time information set at τ_n and becomes public at τ_{n+1} . Innovations are i.i.d. with mean zero and variance $\sigma_{\vartheta}^2 \in (0, \infty)$, independent of the within-interval Brownian shocks. Gaussianity is imposed only when a Gaussian stationary distribution is stated. Two further news channels operate around each reset and are carried inside the reset innovation below. Immediately after τ_n , informed traders receive noisy private signals about the latent innovation, and their transitory positions are absorbed within the interval. At public revelation the basis additionally reprices. These channels are retained because the empirical reset coefficient must be separated from persistent forcing.⁵

⁴OA App. A specifies the interpretation-noise structure, the cancellation condition, and the aggregation when the noise does not cancel exactly.

⁵OA App. A provides the signal-extraction, venue-routing, and reset-auction microfoundations. Appendix B carries the resulting terms through the integration.

3.4 Arbitrageurs and the capital state

Let $y_t \in \mathbb{R}$ denote signed paired notional, with $y_t > 0$ meaning short perpetual and long spot. The real-valued function μ_A is arbitrageurs' perceived per-unit-time convergence return. It is odd, continuously differentiable, satisfies $\mu_A(0) = 0$, and has local slope $\mu'_A(0) = \beta_A > 0$. Conditional on reset-date capital W_n , the sector solves the instantaneous certainty-equivalent problem

$$\max_{y \in \mathbb{R}} \left\{ y(\mu_A(x_t) + c_n) - \frac{y^2}{2\eta_A W_n} \right\} \quad \text{s.t.} \quad |y| \leq \bar{\ell} W_n, \quad (5)$$

where $\eta_A > 0$ scales risk-bearing capacity and $\bar{\ell} > 0$ is maximum paired notional per unit of capital. The exact constrained policy is the projection stated in Section 5.1. While the bound is slack,

$$y_t = \eta_A W_n [\mu_A(x_t) + c_n] = \eta_A W_n (\beta_A x_t + c_n) + o(x_t).$$

Perpetual arbitrage demand is $q_t^A = -y_t$ and spot demand is $+y_t$. The first equality is the exact interior policy under the stated perceived-return map. The second is its local linearization.

A larger W_n allows arbitrageurs to absorb more of both the basis wedge and the funding-induced carry. Section 4 endogenizes W_n through realized profit and liquidation, and Section 5.1 returns to the leverage-saturated branch.

3.5 Liquidity supply, market clearing, and price dynamics

Competitive venue intermediaries absorb the net customer imbalances: retail plus arbitrage demand in the perpetual, the arbitrage hedge in spot, plus the transitory informed flows. Let $\mathcal{H}_t^F$ and $\mathcal{H}_t^S$ denote their perpetual and spot inventories, and let ϖ_t^F and ϖ_t^S denote the expected log-price drift concessions per unit time. With depths $\lambda_F, \lambda_S > 0$ and basis-inventory loadings $\phi_F \geq 0, \phi_S > 0$, the two competitive intermediary problems are

$$\max_{\mathcal{H}^F} \left\{ -\varpi_t^F \mathcal{H}^F - \frac{(\mathcal{H}^F)^2}{2\lambda_F} - \phi_F x_t \mathcal{H}^F - c_n \mathcal{H}^F \right\}, \quad \max_{\mathcal{H}^S} \left\{ -\varpi_t^S \mathcal{H}^S - \frac{(\mathcal{H}^S)^2}{2\lambda_S} + \phi_S x_t \mathcal{H}^S \right\}, \quad (6)$$

where $-c_n \mathcal{H}^F$ is the benchmark carry the perpetual inventory holder pays or receives. Define the inverse depths $\alpha_F \equiv 1/\lambda_F$ and $\alpha_S \equiv 1/\lambda_S$, with $\alpha_A \equiv \alpha_F + \alpha_S$ and $\phi_x \equiv \phi_F + \phi_S$. Market clearing

and the intermediary first-order conditions deliver the two venue-price equations, and subtracting them cancels the common valuation component.⁶ On the slack, local-linear branch the exact basis law is

$$dx_t = (m(W_n)x_t + \psi z_n - d(W_n)c_n - \chi_I e^{-\rho_I(t-\tau_n)} q_n^I) dt + \sigma_x dB_t, \quad t \in [\tau_n, \tau_{n+1}), \quad (7)$$

where B_t is a standard basis Brownian motion, $\sigma_x > 0$ is log-basis volatility per square-root time, the Brownian innovation is independent of reset innovations, and the third drift term is the decaying imprint of transitory informed order flow.⁷ The equilibrium coefficient maps are

$$\begin{aligned} m(W) &= -\phi_x + \alpha_F \eta_M \beta_M - \alpha_A \eta_A \beta_A W = a - bW, \\ d(W) &= 1 + \alpha_F \eta_M + \alpha_A \eta_A W = 1 + d_0 + d_1 W. \end{aligned} \quad (8)$$

The derived coefficients are $a \equiv -\phi_x + \alpha_F \eta_M \beta_M$, $b \equiv \alpha_A \eta_A \beta_A > 0$, $d_0 \equiv \alpha_F \eta_M > 0$, $d_1 \equiv \alpha_A \eta_A > 0$, and $\psi \equiv \alpha_F \eta_M \gamma$. The forcing loading ψ may take either sign under its primitive definition.

Proposition 1 (Local within-interval equilibrium and the basis law). *Fix (X_n, z_n, W_n) with $W_n > 0$. Under the local linearizations of μ_M and μ_A , and while the arbitrage constraint remains slack, a unique local-linear myopic competitive equilibrium exists up to exit from that branch. Its basis process is (7), with coefficient maps (8). Public and informed news enter additively.*

The coefficient $m(W)$ is untreated within-interval propagation per unit time: it is positive when extrapolative demand dominates inventory resiliency and capital-backed convergence trading. The coefficient $d(W) > 0$ is funding transmission.⁸ Capital stabilizes on both margins, $m'(W) = -b < 0$ and $d'(W) = d_1 > 0$, through the single primitive $\alpha_A \eta_A$: more capital strengthens intrinsic mean reversion and the market response to anticipated funding simultaneously. In the deep-market limit, price-impact and demand-side coefficients vanish, so $m(W)$ falls to zero and $d(W)$ to one, and integrating (7) under (2) gives $X_{n+1} = (1 - \kappa^m)X_n + \varepsilon_{n+1}$: a unit model gain eliminates the inherited

⁶Appendix B states the clearing conditions, solves for the drift concessions, and embeds the resulting price impacts around the common valuation process in the venue-price equations.

⁷The informed exposure loads on the coming innovation as $q_n^I = \eta_I \vartheta_{n+1}$, with $\eta_I > 0$ aggregate informed capacity. It is routed across venues with net price-impact loading $\chi_I \equiv \alpha_S(1 - \theta_I) - \alpha_F \theta_I$ for routing share $\theta_I \in [0, 1]$ and decays at absorption rate $\rho_I \geq 0$. OA App. A gives the signal-extraction microfoundation.

⁸Its leading one is the cash transfer priced by the marginal inventory holder. The remaining terms are behavioral demand responses.

snapshot dislocation in this frictionless benchmark.

3.6 From the continuous-time equilibrium to reset dynamics

The within-interval equilibrium becomes empirically useful only after it is mapped into reset-time dynamics. This subsection and the next derive the structural reset coefficient and characterize the stable and sign-reversing regions of the feedback gain. The measurement of both objects is taken up in Section 6.

Because W_n , z_n , and the benchmark carry c_n are fixed within an interval, the local basis law (7) can be integrated exactly until the state exits the slack, local-linear branch. We distinguish the pre-announcement state $\hat{X}_{n+1} \equiv x_{\tau_{n+1}^-}$ from the post-announcement state $X_{n+1} \equiv x_{\tau_{n+1}}$. The hat denotes before public revelation, not a forecast. The pre-announcement state determines interval profit and the structural margin check and obeys

$$\hat{X}_{n+1} = A(W_n, \kappa^m)X_n + \Psi(W_n)z_n + \hat{\varepsilon}_{n+1}. \quad (9)$$

The three deterministic loadings are

$$\begin{aligned} A(W, \kappa^m) &= e^{m(W)\Delta} - \frac{d(W)\kappa^m}{\Delta} \frac{e^{m(W)\Delta} - 1}{m(W)}, \\ \Psi(W) &= \psi \frac{e^{m(W)\Delta} - 1}{m(W)}. \end{aligned} \quad (10)$$

Only A carries the feedback rule: it is the dimensionless survival coefficient on a dislocation that is predetermined at the reset, while Ψ converts the public state into reset-basis units. The innovation $\hat{\varepsilon}_{n+1}$ combines the propagated Brownian component ε_{n+1}^B with the decaying imprint of the transitory informed flow. It is conditionally mean zero given public information at τ_n and realized by τ_{n+1}^- .⁹

At the reset, public revelation reprices the basis directly and reveals the innovation whose absorbed informed imprint already sits in $\hat{\varepsilon}_{n+1}$. The two channels combine into a net news loading

⁹Appendix B carries both components through the integration in closed form. All expressions use continuous limits at $m(W) = 0$. In particular, $A = 1 - d(W)\kappa^m$ and $\Psi = \psi\Delta$ at that point.

$\Gamma_I(W)$ on ϑ_{n+1} , giving the observed recursion

$$X_{n+1} = A(W_n, \kappa^m)X_n + \Psi(W_n)z_n + \varepsilon_{n+1}, \quad \varepsilon_{n+1} \equiv \Gamma_I(W_n)\vartheta_{n+1} + \varepsilon_{n+1}^B. \quad (11)$$

Conditional on the reset state, ε_{n+1} has mean zero.¹⁰ The homogeneous coefficient A governs local correction and instability, whereas public forcing and venue-level information transmission govern covariance and innovation variance. A raw AR coefficient therefore need not equal A . Section 6 derives the corresponding measurement equation.

3.7 The stability band and flip geometry

Proposition 2 (Stability band and flip geometry). *Fix $W > 0$ on the interior arbitrage branch.*

(i) $A(W, \kappa^m)$ is continuous and strictly decreasing in κ^m , with $A(W, 0) = e^{m(W)\Delta}$.

(ii) When $m(W) > 0$, the economically distinct thresholds are

$$\begin{aligned} \kappa_{\min}(W) &= \frac{m(W)\Delta}{d(W)}, \\ \kappa_{\text{flip}}(W) &= \frac{m(W)\Delta e^{m(W)\Delta}}{d(W)(e^{m(W)\Delta} - 1)}, \\ \kappa_{\max}(W) &= \frac{m(W)\Delta e^{m(W)\Delta} + 1}{d(W) e^{m(W)\Delta} - 1}. \end{aligned} \quad (12)$$

The homogeneous reset dynamics are stable exactly for $\kappa^m \in (\kappa_{\min}, \kappa_{\max})$, and $\kappa_{\min} < \kappa_{\text{flip}} < \kappa_{\max}$. If $m(W) \leq 0$, no strictly positive lower threshold is required.

(iii) *Within the stable region, $A > 0$ below κ_{flip} and $A < 0$ above it. Thus the latter region produces damped sign reversals. Gains above $\kappa_{\max}$ produce explosive reversals.*

The two stability boundaries have different economics. When $m(W) > 0$, extrapolative retail demand makes an untreated dislocation grow during the interval, so a gain below $\kappa_{\min}$ does not offset that propagation. At the upper boundary, discrete feedback acts on a stale state and repeatedly

¹⁰OA App. A constructs the announcement and absorption components of $\Gamma_I(W)$. Appendix B records its closed form.

removes more than the surviving dislocation. The flip threshold lies between these two failures: it marks overshooting without instability. Figure 3 displays this geometry and representative paths.¹¹

The flip threshold itself satisfies $d(W)\kappa_{\text{flip}} = ue^u/(e^u - 1)$, which tends to one as $u = m(W)\Delta$ vanishes. In a deep market, therefore, a composite gain of exactly one places the market at the flip margin. The relevant gain in that statement is the equilibrium response κ^m , and Section 6.1 derives its relation to the exchange template slope κ^{tpl} .

4 Endogenous arbitrage capital, liquidation, and fragility

The preceding analysis freezes arbitrage capital within one funding interval. We now close the model by making that stock the outcome of trading profit and margin-driven liquidation.

4.1 Profit, margin usage, and the capital state

Let Π_t^A denote cumulative marked-to-market profit on the signed paired trade. For a predictable position y_t , profit satisfies¹²

$$\begin{aligned} d\Pi_t^A &= y_t(c_n dt - dx_t) - \frac{\zeta}{2W_n} y_t^2 dt, \quad 0 < \zeta \leq 1/\eta_A, \\ \Pi_n &= \int_{\tau_n}^{\tau_{n+1}} d\Pi_t^A, \quad \tilde{W}_{n+1} = W_n + \Pi_n. \end{aligned} \tag{13}$$

The first term is funding income, the second is the marked change in the basis against the signed convergence position, and the last is a quadratic financing and risk-bearing charge.¹³ The book is chosen using the perceived convergence map μ_A , whereas profit accrues under the realized equilibrium law (7). This perceived–realized wedge permits capital to erode even when the ex ante position problem is well behaved.

At τ_{n+1}^- , the sector finances its completed path and absorbs directional basis risk. The margin requirement $\mathcal{M}_{n+1} > 0$, measured in capital units, carries a fixed component, rises with the largest

¹¹For $m(W) \leq 0$ the lower endpoint of the band is zero and only the explosive boundary binds. Appendix B states the convention.

¹²For the snapshot benchmark, the anticipated carry c_n accrues continuously as a bookkeeping convention, while literal exchange settlement remains discrete.

¹³The coefficient ζ has the same units as $1/\eta_A$. The restriction on ζ makes the realized financing charge part of the quadratic cost already internalized in (5), with the remainder a nonnegative risk-bearing charge.

absolute position held during the interval, and rises with the squared pre-announcement basis under separate loadings on positive and negative dislocations. That directional asymmetry is what later makes liquidation pressure side-dependent in Corollary 1. Liquidation events fire on the resulting margin shortfall: on the positive-equity region $\tilde{W}_{n+1} > 0$, the capital transition is

$$W_{n+1} = \tilde{W}_{n+1}(1 - \delta)^{L_{n+1}}, \quad \mathbb{E}[L_{n+1} | \hat{X}_{n+1}, W_n] = \Lambda(\hat{X}_{n+1}, W_n), \quad \delta \in (0, 1), \quad (14)$$

where $L_{n+1} \in \mathbb{Z}_+$ is the liquidation-event count and each event removes fraction δ of the remaining capacity. Nonpositive $\tilde{W}_{n+1}$ is an absorbing default state. Because profit and the margin gap are evaluated at τ_{n+1}^- , liquidations are processed on the pre-announcement state and the announcement innovation never enters the margin check. Equations (13) and (14) form the slow state transition coupled to the fast price block. Between liquidation clusters, frozen- W threshold analysis is a useful local approximation. During a cascade, the joint transition must be used.

The intensity is directional in the basis and convex in low capital, with $[x]_{\pm} \equiv \max\{\pm x, 0\}$:

$$\begin{aligned} \Lambda(\hat{X}, W) = \Delta \Big[& \lambda_0 + \lambda_x^+ ([\hat{X}]_+)^2 + \lambda_x^- ([\hat{X}]_-)^2 \\ & + \lambda_W \left(\frac{\bar{M}}{W} \right)^v \Big]. \end{aligned} \quad (15)$$

Here $\lambda_0 \geq 0$ is a baseline arrival rate, the nonnegative coefficients λ_x^+ and λ_x^- are side-specific exposures to the squared pre-announcement basis,¹⁴ and $\lambda_W \geq 0$ scales low-capital exposure through the fixed margin component $\bar{M} \geq 0$ with convexity $v > 1$.¹⁵

¹⁴The side interpretation is institutional: the common valuation component moves long and short marks together and cancels from the basis, so a positive dislocation marks down short-perpetual accounts and a negative dislocation marks down long-perpetual accounts. Accordingly, λ_x^+ is read as short-side and λ_x^- as long-side intensity, and a reversal activates the opposite side. The capital transition uses only the aggregate count L_{n+1} , and the interpretation is by position side, with trader classes remaining aggregated.

¹⁵OA App. C states the margin requirement in closed form and the structural count law behind (14), which is Poisson with intensity increasing and convex in the proportional shortfall $(\mathcal{M}_{n+1} - \tilde{W}_{n+1})_+ / \tilde{W}_{n+1}$. It derives (15) as its local separable approximation with the coefficient maps, documents the simulation algorithm, and reports the mechanism checks and their limits.

4.2 Capital fragility, the signed reset coefficient, and local infeasibility

With the capital transition in place, we can ask how the stability band of Section 3.7 moves when capital itself changes. That question has a sharp answer only where untreated dislocations propagate, a restriction substantive enough to impose explicitly.

Assumption 1 (Stressed region). *The fragility and infeasibility results are stated for $W > 0$ satisfying $m(W) = a - bW > 0$.*

In this region extrapolative demand dominates intrinsic resiliency and capital-backed convergence trading. The overshooting results in Proposition 2 do not require the restriction. For the infeasibility statement, let $\bar{\kappa}^m > 0$ denote a finite feasibility ceiling on the local state-to-carry gain.¹⁶

Proposition 3 (Capital fragility, signed reset dynamics, and local interior-branch infeasibility).

Maintain Assumption 1 and write $u = m(W)\Delta$ and $h(u) = (e^u - 1)/u$.

(i) *The minimum stabilizing gain falls strictly with capital:*

$$\kappa'_{\min}(W) = -\Delta \frac{b(1+d_0) + ad_1}{(1+d_0+d_1W)^2} < 0. \quad (16)$$

Consequently, a liquidation from W to $(1-\delta)W$ raises the gain required for stability: $\kappa_{\min}((1-\delta)W) > \kappa_{\min}(W)$.

(ii) *Holding the remaining primitives fixed, the direct capital effect on the signed reset coefficient is*

$$\frac{\partial A(W, \kappa^m)}{\partial W} = -b\Delta e^u - \kappa^m [d_1 h(u) - b\Delta d(W) h'(u)]. \quad (17)$$

The sign is negative at every $\kappa^m \geq 0$ when $d_1 h(u) \geq b\Delta d(W) h'(u)$, and otherwise exactly for κ^m below a finite threshold.

(iii) *No feasible gain stabilizes the interior branch if and only if $\bar{\kappa}^m \leq \kappa_{\min}(W)$. When $\bar{\kappa}^m < \Delta a / (1+d_0)$, the unique boundary is*

$$W^{\text{fut}} = \frac{\Delta a - \bar{\kappa}^m(1+d_0)}{\bar{\kappa}^m d_1 + \Delta b} > 0, \quad (18)$$

¹⁶The ceiling is a reduced-form bound on the implementable marginal response, summarizing behavioral-response, participation, implementation, or market-transmission limits, and is therefore a different object from the settlement-level cap $\bar{f}^{\text{set}}$, which truncates the payment level.

and local interior-branch infeasibility, or funding futility, occurs exactly when $W \leq W^{\text{fut}}$.

Part (ii) is conditional because capital moves three terms at once: it lowers untreated propagation e'' and raises transmission $d(W)$, both of which lower A , while the induced fall in the propagation multiplier $h(u)$ raises A . The latter two channels scale with κ^m , so the net sign can turn only when feedback is strong. The result concerns signed A , not absolute persistence: on the negative- A side a lower A means a stronger reversal and need not reduce $|A|$ or the half-life.

Part (iii) is where the falling boundary of part (i) meets the feasibility ceiling: the boundary W^{fut} is the unique capital level satisfying $\kappa_{\min}(W^{\text{fut}}) = \bar{\kappa}^m$. Both W and $\bar{\kappa}^m$ are latent, so the boundary is structural. The empirical analysis uses capacity and fragility proxies to study its comparative implications.

Part (ii) is the comparative static behind the empirical $X \times \text{Low}W$ interaction and the capacity cross-section. Figure 4 combines parts (i) and (iii): as capital falls, the lower stability boundary rises past a fixed gain and, at sufficiently low capital, past the maximal feasible gain, leaving no feasible stabilizing choice on the interior branch. Global saturation bounds the resulting amplification.¹⁷

4.3 Sign reversals and directional liquidation

Corollary 1 (Sign reversals activate opposite-side liquidation). *Fix $W_n > 0$ and $X_n \neq 0$, and set all additive forcing terms in (11) to zero. If $\kappa^m > \kappa_{\text{flip}}(W_n)$, then $\text{sgn}(\hat{X}_{n+1}) = -\text{sgn}(X_n)$, and the opposite-side component of (15), $\Delta\lambda_x^\mp A(W_n, \kappa^m)^2 X_n^2$ with $\mp$ opposite to $\text{sgn} X_n$, is zero at $\kappa^m = \kappa_{\text{flip}}(W_n)$ and strictly increasing in κ^m beyond it whenever $\lambda_x^\mp > 0$. If $0 \leq \kappa^m \leq \kappa_{\text{flip}}(W_n)$, the homogeneous component does not activate the opposite-side term.*

Two features of the corollary matter for the empirical design. Because the flip margin lies inside the stable band, opposite-side exposure arises already in the damped-reversal region and does not require instability. And the threshold statement concerns only the homogeneous feedback component: below the flip margin, forcing and innovations can still generate realized reversals, which is why the empirical definition of an economic reversal filters small crossings. A generic

¹⁷Retail beliefs saturate while arbitrage exposure and carry are bounded, so their outward contributions cannot grow without limit, whereas intermediary resiliency remains linear in the basis. Appendix B shows that the conditional drift therefore points inward beyond a finite, state-dependent radius. That result bounds drift: a stationary joint distribution would additionally require regularity and moment restrictions on the forcing and capital processes.

volatility shock can raise both sides' liquidations, whereas the feedback mechanism predicts which side becomes exposed after the reversal. This directional prediction motivates the same-side placebo design in Section 7.4.

4.4 The coupled basis–capital mechanism

Combining the reset law with the capital transition closes the feedback loop. A dislocation changes trading profit and margin demand. Liquidation reduces capital. Lower capital raises the minimum stabilizing gain. The next dislocation is therefore absorbed by a weaker arbitrage sector. Retained convergence profit and outside entry work in the opposite direction. Proposition 3 moves the stability boundary after an exogenous change in capital. We now endogenize that change through liquidation in a deterministic mean-field benchmark that also omits outside entry.¹⁸

Proposition 4 (Liquidation ratchet and single-burst entry into infeasibility). *Consider the deterministic mean-field benchmark with zero forcing and $\Pi_n = 0$, evolving according to*

$$\begin{aligned} X_{n+1} &= A(W_n, \kappa^m) X_n, \\ \Lambda_n &= \Lambda(A(W_n, \kappa^m) X_n, W_n), \\ W_{n+1} &= W_n (1 - \delta)^{\Lambda_n}. \end{aligned} \tag{19}$$

Maintain the stressed region and $\bar{\kappa}^m < \Delta a / (1 + d_0)$.

(i) *Whenever $\Lambda_n > 0$, capital falls and the minimum stabilizing gain rises: $W_{n+1} < W_n$ and $\kappa_{\min}(W_{n+1}) > \kappa_{\min}(W_n)$.*

(ii) *If $W_0 > W^{\text{fut}}$ and $\Lambda_0 \geq \log(W^{\text{fut}}/W_0)/\log(1 - \delta)$, then $W_1 \leq W^{\text{fut}}$.*

Part (ii) says that a single sufficiently large mean-field liquidation burst moves the interior branch into the conditional infeasibility region. The required mean count is approximately $\log(W_0/W^{\text{fut}})/\delta$ for small δ . Part (ii) is a one-step threshold-crossing result, not a duration model. The numerical checks below restore the suppressed ingredients and show which conclusions survive.¹⁹

¹⁸Because the capital transition is nonlinear in the count, the plug-in path is not the conditional expectation of the stochastic transition. It traces threshold crossing.

¹⁹The integer-jump counterpart and its smallest crossing count are recorded in OA App. C.

Numerical mechanism checks. Because the analytical results isolate local derivatives at a frozen capital state, we simulate the global nonlinear system, with saturated beliefs, clipped positions, endogenous branch switching, realized profit, structural margin gaps, and random liquidation counts. The simulations confirm that the threshold logic survives these nonlinearities under the illustrative calibration: below the illustrative infeasibility boundary the local linear coefficient exceeds one while the full paths are pulled inward in the remote tails, so Proposition 3(iii) and the tail-drift qualification are compatible. They also show that recovery from low-capital states is highly sensitive to outside capital entry. Under the illustrative calibration, simulated residence below the boundary is much longer than empirical cap spells, so the exercise is a mechanism check, not a quantitative duration fit.²⁰

5 Implementation constraints and design

The interior model isolates the capital-dependent elasticity that makes funding effective. Two constraints remove that elasticity in different places. The arbitrage leverage bound eliminates the marginal response of convergence traders. The exchange clamp and rate cap eliminate the marginal response of the payment rule. The exchange cap is directly observable, whereas private leverage saturation is latent, so the two constraints enter the empirical analysis differently. The first two subsections study these constraints. The third studies the two design margins that reshape feedback: the funding interval, which sets how often the rule acts, and the gain, which governs the persistence and stationary risk a given capital state inherits. For any lower and upper bounds $\ell \leq u$, write $\text{clip}(r, \ell, u) \equiv \min\{u, \max\{\ell, r\}\}$ for Euclidean projection onto $[\ell, u]$.

5.1 When the arbitrage constraint binds

The exact arbitrage policy implied by (5) is

$$y_t = W_n \text{clip}(\eta_A[\mu_A(x_t) + c_n], -\bar{\ell}, \bar{\ell}). \quad (20)$$

²⁰OA App. C reports the algorithm, the illustrative calibration, and the full duration and sensitivity diagnostics. Under the strict capital-transition formula, in which economic default requires pre-liquidation equity at or below zero, no simulated path defaults at any entry speed. Apparent default fractions under a positive numerical absorption threshold measure absorption at that threshold.

On the interior branch, the arbitrage position responds at the margin to both the basis and anticipated carry. On a strict fixed-sign leverage corner, the position remains in the market but is locally constant, so both marginal responses vanish. Consequently, the capital-backed terms disappear from both coefficient maps: $m(W) = a - bW$ becomes a , while $d(W) = 1 + d_0 + d_1W$ becomes $1 + d_0$. The constraint therefore changes both untreated basis propagation and carry transmission discontinuously. The following result conditions on the same leverage boundary remaining active with the same sign throughout the interval.

Proposition 5 (Binding constraints and a second route to local infeasibility). *Suppose $y_t = \varsigma_n \bar{\ell} W_n$ with fixed $\varsigma_n \in \{-1, +1\}$ throughout an interval. The reset recursion is affine, with homogeneous coefficient*

$$A^{\text{bind}}(\kappa^m) = e^{a\Delta} - \frac{(1 + d_0)\kappa^m}{\Delta} \frac{e^{a\Delta} - 1}{a}, \quad (21)$$

using its continuous limit at $a = 0$. If $a > 0$, write $\kappa_{\min}^{\text{bind}} \equiv a\Delta/(1 + d_0)$. The fixed-sign branch is locally stable precisely when

$$\frac{a\Delta}{1 + d_0} < \kappa^m < \frac{a\Delta}{1 + d_0} \frac{e^{a\Delta} + 1}{e^{a\Delta} - 1}. \quad (22)$$

Moreover, whenever the interior branch at the same W is stressed, $\kappa_{\min}^{\text{bind}} > \kappa_{\min}(W)$. Thus an interior state can be stabilizable even though the leverage-saturated branch is not.

The economic content is simpler than the algebra.²¹ Low W weakens arbitrage continuously on the interior branch. A binding leverage constraint removes marginal arbitrage elasticity discretely. A chosen gain $\kappa^m \leq \kappa_{\min}^{\text{bind}}$ makes the saturated branch locally unstable, but that alone is not infeasibility because a larger feasible gain may exist. The binding branch creates a second route to local infeasibility only when the maximal feasible gain $\bar{\kappa}^m$ lies weakly below $\kappa_{\min}^{\text{bind}}$. A full path switches endogenously among the interior, leverage-bound, and funding-cap regions.²² Figure 5 illustrates why A^{bind} is a branch coefficient: a global path can start with the leverage bound and the carry cap jointly binding, leave the leverage corner while the cap still binds, and later return to the interior.

²¹Appendix B gives the complete affine recursion, its forcing and inventory level shifts, and the algebra behind the strict threshold gap.

²²OA App. D studies the branch-validity conditions and switching paths.

The funding-rate cap is directly observable in the data, whereas the private leverage state in (20) is latent. Proposition 5 therefore serves as a microfoundation for the constrained branch.

5.2 The literal exchange rule and a normalized snapshot benchmark

We use three related but non-equivalent objects. Equation (1), completed by the mechanical map (23) below, records how the literal exchange rule computes a settled transfer from the interval-average premium. Section 6.1 studies the conditional-expectations fixed point on that rule's active, uncapped same-interval branch. The normalized snapshot rule introduced below is a parallel benchmark that isolates the local derivatives of the flat, active, and capped branches. It complements the timing correction in Section 6.1. Empirical clamp and cap regions are assigned using the literal rule.

Using this clipping notation, the literal Binance-style settlement map is

$$f_{n+1}^{\text{set}} = G(\bar{P}_n) = \text{clip}\left(\mathcal{T}(\kappa^{\text{tpl}}\bar{P}_n), -\bar{f}^{\text{set}}, \bar{f}^{\text{set}}\right), \quad \mathcal{T}(p) \equiv p + \text{clip}(\iota - p, -\delta^{\text{cl}}, \delta^{\text{cl}}). \quad (23)$$

Here $p = \kappa^{\text{tpl}}\bar{P}_n$ is the uncapped template input, $\kappa^{\text{tpl}} > 0$ is the nominal active-branch derivative, $\iota \in \mathbb{R}$ is the interest-component transfer, $\delta^{\text{cl}} > 0$ is the clamp half-width in transfer units, and $\bar{f}^{\text{set}} > 0$ is the absolute interval-settlement cap.²³

Before the outer cap, $\mathcal{T}$ shifts its input by $\pm\delta^{\text{cl}}$ outside the interest band and pins it at ι within $[\iota - \delta^{\text{cl}}, \iota + \delta^{\text{cl}}]$, so its derivative is one, zero, and one across those three branches. Under the maintained institutional ordering $|\iota| < \bar{f}^{\text{set}}$ the outer cap adds two further zero-derivative regions, and dividing the four resulting input-space kinks by κ^{tpl} locates them in $\bar{P}$ units. Accordingly, the local derivative of G with respect to $\bar{P}$ is zero in the flat and capped regions and κ^{tpl} on either active, uncapped branch. Equation (23) is the mechanical settlement rule conditional on the realized $\bar{P}_n$, and the equilibrium basis response is derived separately.

For closed-form branch mechanics, the normalized benchmark removes the interest intercept, replaces the interval-average input by the reset state, and expresses the cap in snapshot-carry units, isolating the qualitative zero/positive/zero sequence of local payment slopes. Equation (24) is a derivative-pattern benchmark. After removing the interest intercept, zero in the dead zone denotes zero incremental basis-responsive carry, while literal exchange funding remains at the interest

²³The notation δ^{cl} avoids confusing the interest clamp with the informed-sector superscript I .

component. Let $x_{\text{dz}} > 0$ be the resulting symmetric log-basis half-width and let $\bar{c} > 0$ be a cap on the per-unit-time snapshot carry. The piecewise analysis also requires a strictly positive gain, $\kappa^m > 0$: at $\kappa^m = 0$ the map degenerates to $g \equiv 0$, and no active or capped branch exists. The normalized map sends X_n into the total anticipated interval transfer Δc_n :

$$g_{x_{\text{dz}}, \bar{c}}(x) = \begin{cases} 0, & |x| \leq x_{\text{dz}}, \\ \kappa^m(x - x_{\text{dz}} \text{sgn} x), & x_{\text{dz}} < |x| \leq x_{\text{dz}} + \Delta \bar{c} / \kappa^m, \\ \Delta \bar{c} \text{sgn} x, & |x| > x_{\text{dz}} + \Delta \bar{c} / \kappa^m, \end{cases} \quad c_n = \frac{g_{x_{\text{dz}}, \bar{c}}(X_n)}{\Delta}. \quad (24)$$

For $m(W) \neq 0$, define the dimensionless transmission factor $H(W) \equiv d(W)[e^{m(W)\Delta} - 1]/[\Delta m(W)]$, with continuous value $H(W) = d(W)$ at $m(W) = 0$: it is the reset-date basis effect of one unit of total anticipated interval carry, a translation factor.

Proposition 6 (Piecewise reset dynamics under the normalized clamp and cap). *Under (24), the reset recursion and its local state derivative away from the four kink points (two symmetric kink pairs) are*

$$\begin{aligned} X_{n+1} &= e^{m(W_n)\Delta} X_n + \Psi(W_n) z_n - H(W_n) g_{x_{\text{dz}}, \bar{c}}(X_n) + \varepsilon_{n+1}, \\ \frac{\partial X_{n+1}}{\partial X_n} &= \begin{cases} e^{m(W_n)\Delta}, & |X_n| < x_{\text{dz}}, \\ A(W_n, \kappa^m), & x_{\text{dz}} < |X_n| < x_{\text{dz}} + \Delta \bar{c} / \kappa^m, \\ e^{m(W_n)\Delta}, & |X_n| > x_{\text{dz}} + \Delta \bar{c} / \kappa^m. \end{cases} \end{aligned} \quad (25)$$

The dead zone and the cap have very different payment levels but the same marginal law. A capped payment can be large while providing no additional response to a marginal increase in the basis. The centered dead-zone payment is instead small. On both branches the payment is locally constant in the reset state, so an additional unit of basis receives no additional funding correction and propagates at the untreated rate $e^{m(W)\Delta}$. Only the intermediate branch retains the model gain and therefore the coefficient $A(W, \kappa^m)$. When $m(W) > 0$ and the active-branch gain lies in the stability band, the rule can stabilize intermediate dislocations while leaving both very small and very large dislocations locally unanchored. If $m(W) < 0$, the zero-slope regions remain mean reverting but lose the additional pull from funding.²⁴

²⁴The full branch-by-branch level recursion is reported in Appendix B.

The empirical analysis assigns regions using the actual asymmetric, interest-centered template (23) applied to $\bar{P}_n$. Proposition 6 carries the literal rule's qualitative derivative pattern into a reset-state benchmark. Connecting its branch coefficients to observed reset persistence and to literal clamp or cap states requires the timing and measurement conditions in Section 6.

5.3 The funding interval and the gain schedule

Institutionally, the funding interval governs both the horizon over which the premium input is averaged and the frequency at which the resulting transfer is settled. In the snapshot benchmark, Δ instead denotes the time between reset states and the horizon over which the fixed carry c_n accrues. Write $A(W, \kappa^m; \Delta)$ to make that benchmark interval explicit.

Two interval experiments. We separate two design questions. First, suppose the exchange retunes the gain as Δ changes so that each reset retains a fixed feasible target coefficient $\bar{A}$. Second, suppose instead that Δ changes while the per-interval model gain κ^m is held fixed. The first experiment holds the effect of each reset constant. The second holds the total anticipated transfer for a given reset basis constant and applies that intervention more or less frequently in calendar time.

Corollary 2 (Funding interval and calendar-time persistence). *Fix $W > 0$ and $0 < |A(W, \kappa^m; \Delta)| < 1$.*

- (i) *The homogeneous half-life is $\log(1/2)/\log |A(W, \kappa^m; \Delta)|$ funding steps, and Δ times that in calendar time.*
- (ii) *If the gain is retuned across interval lengths so that $A(W, \kappa^m; \Delta) = \bar{A}$ for a fixed attainable $\bar{A} \in (-1, 1)$, $\bar{A} \neq 0$, the calendar half-life is $\Delta \log(1/2)/\log |\bar{A}|$.*
- (iii) *At fixed κ^m , $\partial A/\partial \Delta > 0$ throughout the stability band when $m(W) > 0$. When $m(W) < 0$, the derivative can take either sign.*

Under the stressed-market case $m(W) > 0$, a shorter interval at fixed κ^m lowers the signed coefficient A . This is stabilizing while $A > 0$ because it lowers $|A|$. Once $A < 0$, a further decline makes A more negative, raises $|A|$, enlarges the alternating correction, and moves the system toward the $A = -1$ boundary. For example, a move from 0.8 to 0.4 speeds monotone decay, whereas a move from -0.3 to -0.7 strengthens reversal and slows absolute decay. When $m(W) < 0$, untreated market forces are already mean reverting: a longer interval gives that intrinsic convergence more

time to operate but also changes the cumulative transmission of funding, so the two effects need not share a sign.

Changing Δ alters sampling, the transfer per interval, and possibly the exchange's endogenous response to stress. Because interval switches in the data follow extreme funding states, Section 7.6 analyzes them descriptively. Corollary 2 supplies the fixed- W , fixed-gain comparative static used to organize those patterns.

The interval is one design margin. The gain is the other. Stability is a first-moment property: on the interior branch the scalar stationary-variance amplification factor is $1/(1 - A^2)$, which diverges as A approaches either 1 or -1 , so too little feedback leaves the process near a positive unit root and too much leaves it near a negative unit root, and a coefficient inside the band is not by itself a low-risk configuration. Because $A(W, \kappa^m)$ is affine and strictly decreasing in κ^m by Proposition 2(i), any feasible interior target is attained by a unique nonnegative gain, and the target $\bar{A} = 0$ reproduces $\kappa_{\text{flip}}(W)$ by definition. The coefficient is capital dependent, however, so a fixed gain lets it drift toward either boundary as W moves: holding a chosen target requires a gain schedule, which a flat region, rate cap, feasible-gain bound, or binding arbitrage constraint can nonetheless place out of reach.²⁵

6 From theory to empirical objects

This section maps the model's local feedback objects into the settlement rates, interval-average premiums, and reset coefficients observed in the data. The mapping is reduced form because arbitrage capital and the persistent forcing state remain latent.

6.1 Settlement timing and the three funding slopes

For contract i and interval n , let $f_{i,n}^{8,\text{fwd}}$ denote the total rate settled at τ_{n+1} from premium observations during $[\tau_n, \tau_{n+1})$, normalized to an eight-hour unit. It is the realized transfer associated with carrying

²⁵OA App. B states and proves the stationary-risk result, including the divergence of the basis variance at both boundaries when $m(W) > 0$, the minimization of the scalar amplification component at the flip gain, and Gaussianity under Gaussian innovations. OA App. G states the gain schedule in closed form, records which targets deliver monotone decay and which deliver damped reversals, and shows that at capital states where a fixed gain leaves $|A| \geq 1$ the schedule delivers a finite local stationary second moment where the fixed rule does not.

a position through interval n , but it is not generally known at τ_n . For exercises whose outcomes are measured over $[\tau_n, \tau_{n+1})$, the intended lagged-settlement indexing is

$$f_{i,n}^8 \equiv f_{i,n-1}^{8,\text{fwd}}.$$

Under this indexing, the rate was already settled at τ_n from observations in the preceding interval and is predetermined for interval- n outcomes. The event-time distinction is essential for the liquidation design, and the liquidation exercises use it: funding stamped at τ_n is related to liquidation over the following interval. Predetermination here refers to timing: the settled rate remains endogenous to the recent basis from which it is computed. The theoretical c_n in (2) is instead a precommitted or conditionally anticipated carry flow, whereas the two empirical rates are realized settlement measures. Let $f^{\text{tpl}}(\bar{P}) \equiv G(\bar{P})$ denote the published institutional map in (23).

For each contract, Cov_i and Var_i denote within-contract population moments, with $\text{Var}_i(X_{i,n}) > 0$. The empirical counterparts below are estimated on a common contract-level sample under a common residualization convention. The three funding slopes are

$$\begin{aligned} \kappa^{\text{tpl}} &\equiv \left. \frac{\partial f^{\text{tpl}}(\bar{P})}{\partial \bar{P}} \right|_{\text{active, uncapped branch}}, \\ r_i^{\text{int}} &\equiv \frac{\text{Cov}_i(\bar{P}_{i,n}, X_{i,n})}{\text{Var}_i(X_{i,n})}, \\ \kappa_i^{\text{eff}} &\equiv \frac{\text{Cov}_i(f_{i,n}^{8,\text{fwd}}, X_{i,n})}{\text{Var}_i(X_{i,n})}. \end{aligned} \tag{26}$$

The first is the nominal institutional derivative on an active, uncapped branch. The second is an equilibrium interval-retention projection. The third is the reduced-form slope of the forward-settled payment on the reset-time state.

These projections have a structural counterpart, because settlement on the same interval creates a timing fixed point. A larger anticipated payment induces traders and arbitrageurs to compress the basis earlier in the interval, but that earlier compression lowers the average premium on which the payment is calculated, so the realized rule attenuates its own state-to-payment slope. The precommitted snapshot benchmark contains no such attenuation.

We isolate the fixed point on one active, uncapped branch of G , writing the centered exchange

input locally as $P_t = \lambda_P x_t$ with premium-index loading $\lambda_P > 0$.²⁶ Writing $\mathbb{E}_n[\cdot]$ for expectations conditional on public information immediately after τ_n , the branch transfer is $f_{n+1}^{\text{act}} = \kappa^{\text{tpl}} \bar{P}_n$ and the anticipated carry closes the loop through $c_n = \mathbb{E}_n[f_{n+1}^{\text{act}}]/\Delta$. With the dimensionless propagation index $u \equiv m(W)\Delta$ and the averaging functions $h(u) = (e^u - 1)/u$ and $g(u) = (e^u - 1 - u)/u^2$, where h propagates the opening basis into its interval average and g accumulates a constant carry flow into that average, solving the fixed point gives the structural state-to-payment slope

$$\frac{\partial \mathbb{E}_n[f_{n+1}^{\text{act}}]}{\partial X_n} = \frac{\kappa^{\text{tpl}} \lambda_P h(u)}{1 + d(W) \kappa^{\text{tpl}} \lambda_P g(u)}. \quad (27)$$

The denominator is the endogenous attenuation, and it is stronger the larger the market's carry transmission $d(W)$: the effective state-to-payment slope is generally smaller than the nominal template derivative, so market location is stated in terms of the reset coefficient A .²⁷ Equation (27) is the clean active-branch structural analogue of κ_i^{eff} , and its premium-input factor that of r_i^{int} . Forcing, nonlinear regions, and measurement wedges can separate the empirical projections from these structural counterparts.

Within a locally linear settlement region, let κ_i^{avg} denote the contract-level population projection slope of the settled rate on its own interval-average premium, so that $f_{i,n}^{8,\text{fwd}} = \alpha_i^f + \kappa_i^{\text{avg}} \bar{P}_{i,n} + e_{i,n}^f$ with projection residual $e_{i,n}^f$. Because r_i^{int} is itself the projection slope of $\bar{P}_{i,n}$ on $X_{i,n}$, the three slopes satisfy the exact identity

$$\kappa_i^{\text{eff}} = \kappa_i^{\text{avg}} r_i^{\text{int}} + \frac{\text{Cov}_i(e_{i,n}^f, X_{i,n})}{\text{Var}_i(X_{i,n})}. \quad (28)$$

The multiplicative decomposition $\kappa_i^{\text{eff}} = \kappa_i^{\text{avg}} r_i^{\text{int}}$ therefore holds exactly when the settlement residual is orthogonal to the reset-time state, and $\kappa_i^{\text{avg}} = \kappa^{\text{tpl}}$ when the template is implemented without a measurement or settlement wedge in that region.²⁸ An estimation sample that mixes flat, active, capped, or transition observations can leave that residual correlated with $X_{i,n}$, in which case the effective slope differs from the product by exactly the residual term. Pooled fixed-effects slopes and cross-contract medians are different objects and are therefore reported separately.

²⁶The benchmark is local to one active, uncapped branch, away from the interest-centered flat region and the cap. Variables are deviations from the local branch reference point, with the branch intercept and additive forcing suppressed. The loading $P_t = \lambda_P x_t$ is a local approximation that leaves κ^{tpl} as the nominal derivative of the exchange map.

²⁷Appendix B derives (27). OA App. E records the reset-coefficient counterpart of the same fixed point, its stability boundaries, and their comparison with the snapshot benchmark.

²⁸Appendix B derives (28) and records the general expression for the effective slope outside a locally linear region.

6.2 Measurement of reset dynamics

The empirical counterpart of the reset coefficient A is the slope of X_{n+1} on X_n . Fix W . Because the reset innovation is orthogonal to the predetermined state, the recursion (11) implies the measurement equation

$$\beta^{\text{raw}} \equiv \frac{\text{Cov}(X_{n+1}, X_n)}{\text{Var}(X_n)} = A(W, \kappa^m) + \Psi(W) \frac{\text{Cov}(z_n, X_n)}{\text{Var}(X_n)}, \quad (29)$$

and the cross-covariance $\text{Cov}(z_n, X_n)$ is generically nonzero.²⁹

The raw reset coefficient therefore combines the structural survival of an inherited dislocation with covariance from persistent forcing, and its sign need not reveal the sign of the structural A . Controlling for the true state z_n would recover A under $\mathbb{E}[\varepsilon_{n+1} \mid X_n, z_n, W] = 0$. With a vector of proxy controls p_n , write $\tilde{X}_n \equiv X_n - \text{Proj}(X_n \mid p_n)$. Recovery then requires spanning, $\text{Cov}(\tilde{X}_n, z_n) = 0$, and orthogonality, $\text{Cov}(\tilde{X}_n, \varepsilon_{n+1}) = 0$, and whether a given set of controls satisfies them is an empirical question.

We therefore construct a proxy-adjusted coefficient A_i^{adj} , the contract-level slope of $X_{i,n+1}$ on $X_{i,n}$ after controlling for a trailing own mean and a leave-one-out market mean, and use it as a common residualized measure of sign and cross-contract ordering.³⁰

7 Empirical evidence

The evidence is organized by how close each exercise comes to the model. Section 7.1 describes the panel and delimits its scope. Sections 7.2 and 7.3 measure the state-to-settlement mapping and the residualized reset dynamics, and Sections 7.4 and 7.5 document the directional-liquidation and cap-state footprints the model implies. Section 7.6 evaluates the identification of the model's dynamic comparative statics.

²⁹A lagged public state propagates into both the current basis and the current public state, and the same reset innovation enters both z_n and the announcement component of X_n . Appendix B derives (29) and gives the closed form of the cross-covariance when the triangular system (4)–(11) is covariance stationary.

³⁰Appendix B proves the core theory. OA App. H maps each empirical exercise below to the theoretical object it tests and states the interpretation boundary each one carries, including for the auxiliary and null tests. The rest of the OA contains numerical implementation details, robustness exercises, and additional analytical results.

7.1 Data and empirical scope

The sampling unit is a contract-by-reset observation: one USDT-margined perpetual i at one funding reset n , spanning $[\tau_n, \tau_{n+1})$. Four raw sources are merged at this frequency: funding rates from Binance Vision, including the exact funding-interval field; 5-minute spot and futures klines, from which the reset-time basis and interval realized volatility are built; 5-minute open-interest and positioning metrics; and tick-level liquidations from Tardis. The final panel is 677,405 contract-reset observations on 200 perpetuals over 2020-09 to 2025-12, all on the common 8-hour reset grid. Interval-switch episodes are analyzed separately, on the native funding grid, in Section 7.6.³¹

Because curated tier labels are coarse, contracts are sorted into within-sample quintiles of median open-interest value (quintile 1 thinnest, median OI \$3M; quintile 5 deepest, \$47M). This is a capacity sort motivated by the model, not an observation of W : depth, price impact, trader composition, and effective gain move with it, and a majors-only panel corresponds essentially to the deepest quintile alone. Binance rate-limits its public liquidation stream from April 2021 onward, so the largest cascades are undercounted and the truncation is most severe in exactly the extreme states the model is about. Reported liquidation levels below are therefore lower bounds, and side-specific truncation cannot be ruled out.³² Unless stated otherwise, regressions absorb contract and reset-time fixed effects α_i and δ_n and report standard errors clustered two ways.³³

Tables 2 and 3 summarize the panel. The reset-level basis is small and quickly mean reverting: mean -4.3 bps, median -5.2 bps, standard deviation 9.5 bps, with a 20-reset rolling coefficient averaging 0.11 , a reduced-form property of the joint market and rule. Funding is tightly clustered near the interest component. Cap binding is rare overall (0.083% of resets) and economically meaningful sign reversals occur at 2.8% . The capacity gradients described in Section 2 are visible here in detail, and the capacity contrasts below are read as reduced-form cross-sectional sorts.

³¹From 298 discovered contracts with at least 18 months of funding history and matching data, 200 enter the common eight-hour panel: 5 tier-one majors, 19 mid-caps, and 176 thin altcoins. Most exclusions are four-hour-native contracts without usable leads on that grid. OA App. I reports the full sample-selection accounting, the cutoff sensitivity, and the construction choices that matter in the broad panel.

³²The truncation occurs at the outcome-recording layer but can propagate into regression coefficients because the dependent variables are constructed from recorded levels, with the direction and magnitude of the resulting bias undetermined. A comparison of opposite- and same-side outcomes reduces common undercounting but does not remove side-specific truncation.

³³Clustering on reset time is essential because market-wide moves induce correlated basis shocks across hundreds of contracts simultaneously. Table A.1 defines every constructed variable used below.

7.2 The state-to-settlement mapping

The state-to-settlement mapping sends the basis observed at a reset into the funding rate settled at the close of the interval that reset opens. Contract by contract, we measure its average slope by regressing forward-settled per-8h funding on the lagged reset-time basis. Across 200 contracts the median κ^{eff} is 0.142 (interquartile range [0.099, 0.200]; Figure 1), below the template anchor $\kappa^{\text{tpl}} \approx 1$ under this reset-basis proxy, and it varies little across capacity quintiles. The comparison with the template and the size of the gap depend on which observable series stands in for the state, a sensitivity quantified later in this subsection.³⁴

The compression decomposes institutionally: the clamp acts on an interval-average premium input, and $\bar{P}_n$, the interval mean of the five-minute perpetual-minus-spot basis, stands in for that input. Because Binance samples the premium index far more frequently than every five minutes and weights observations closer to settlement more heavily, the five-minute interval mean is an approximation to the rule’s actual input.³⁵ Two pooled panel regressions separate the two links in the chain,

$$f_{i,n}^{8,\text{fwd}} = \kappa_R^{\text{pool}} \bar{P}_{i,n} + \alpha_i + \delta_n + \varepsilon_{i,n}, \quad \bar{P}_{i,n} = r^{\text{pool}} X_{i,n} + \alpha_i + \delta_n + \eta_{i,n}, \quad (30)$$

the first estimated separately on each clamp region R of $\bar{P}$ and the second on the full estimation sample. Template pass-through is 0.856 below the flat region, -0.009 inside, and 0.253 above, and the pooled interval-retention slope is 0.384 (Figure 2). Contract-level estimates answer a different question (Table 4). They are the sample analogues of the within-contract population projections of Section 6.1, estimated one contract at a time. Median non-flat pass-through is 0.232 , median interval retention is 0.691 , and the median of their contract-level product is 0.161 against median $\kappa^{\text{eff}} = 0.143$, with cross-contract correlation 0.953 . This is a decomposition check under (28), not an identity among pooled medians. Rule pass-through is classified by the interval-average premium proxy, the argument the template actually clamps, whereas the persistence estimates below are indexed by reset-time basis bins aligned with the nominal thresholds. The two exercises use

³⁴The trailing-settled variant has median 0.121 . Quintile medians run $0.126, 0.141, 0.127, 0.123$, and 0.160 from thinnest to deepest, so the capacity ordering is not monotone.

³⁵Binance Clearing, USDS-Margined Futures Clearing Procedures, pp. 24–26.

different region assignments, so their region-specific coefficients answer different questions.³⁶

To gauge proxy sensitivity, we re-estimate the first regression in (30) on a second observable series, the interval mean of five-minute premium-index klines. That series is closer to the documented settlement input than the basis proxy. On the 669,688 resets and 190 contracts where both series exist their median cross-contract correlation is 0.941, yet the estimates they deliver differ materially: pass-through is 1.209 below the flat region, 0.030 inside, and 0.879 above, so outside-region pass-through is near the template value of one under the closer proxy. A companion projection of settled funding on the premium-index state itself gives a median slope of 0.422, against the median κ^{eff} of 0.143 on the reset-time basis. Because the two projections use different state variables, they measure different state-to-payment mappings, and their gap shows that the measured attenuation is sensitive to how the exchange input is approximated. Both series deliver the qualitative pattern the paper relies on, a flat central region with positive slope on either side.³⁷

The reduced-form funding–basis slope is economically large below the flat region (+0.37, $t = 8.0$), nearly zero inside it (+0.012, $t = 4.0$), and small above it (+0.015, $t = 1.8$). These estimates document substantial branch heterogeneity and strong attenuation inside and above the flat region, with estimated persistence between 0.17 and 0.34 across region–fragility cells, well below the amplification threshold of one. A local kink design at the clamp folds is inconclusive: the folds lie only ten basis points apart, a placebo fold at -24 basis points is non-null, and the positive-side placebos have too little support to test. We therefore use the region assignments to summarize branch heterogeneity.³⁸

The region-by-fragility ordering also differs from the simple branch comparison implied by Proposition 6 (the Online Appendix table on region-by-fragility persistence). Because both the region labels and the reset coefficients are reduced-form objects that mix settlement timing, persistent forcing, and measurement components, we treat these cells as a diagnostic of branch heterogeneity.

³⁶Under the $\bar{P}$ assignment the region shares are 65.7% below, 21.7% inside, and 12.6% above the flat region. Using the reset-time basis instead, the three regions contain 394,955 (58.3%), 191,044 (28.2%), and 91,406 (13.5%) observations. The shares differ, which is why region-specific coefficients from the two exercises are not directly comparable. No arithmetic decomposition uses either set of shares.

³⁷The two series also imply different pooled reset persistence, 0.315 on the reset basis against 0.754 on the interval-mean premium-index proxy. They use different temporal aggregation, so that gap is a measurement diagnostic.

³⁸OA App. I reports the region-by-fragility persistence cells and the kink-design estimates and figure.

7.3 Reset dynamics and cross-contract heterogeneity

The raw full-sample contract-level slope of X_{n+1} on X_n has median 0.652. Slow-moving premium components, such as market-wide news and contract-specific carry regimes of the type highlighted by the measurement equation (29), can inflate this coefficient, so we estimate, one contract at a time,

$$X_{i,n+1} = a_i + A_i^{\text{adj}} X_{i,n} + b_i \bar{X}_{i,n}^{\text{own}} + c_i \bar{X}_{-i,n}^{\text{mkt}} + u_{i,n+1}, \quad (31)$$

where $\bar{X}_{i,n}^{\text{own}}$ is the contract's trailing 90-reset mean basis, a carry-regime proxy, and $\bar{X}_{-i,n}^{\text{mkt}}$ is the leave-one-out cross-sectional mean basis at reset n , a market-news proxy. These two variables are the proxy controls p_n of Section 6. Replacing the dependent variable by $f_{i,n}^{8,\text{fwd}}$ and keeping the same controls defines the proxy-adjusted funding slope $\kappa_i^{\text{eff,adj}}$.³⁹

Three reduced-form findings emerge (Table 5 and Figure 6). First, every contract has a positive adjusted point estimate under the adopted proxy controls. Second, the median falls from 0.652 raw to 0.242 adjusted, so slow common and contract components materially affect the observed recursion. Third, the endpoint contrast is positive: the thinnest quintile has a lower median coefficient (0.200) than the deepest (0.275), although the interior quintiles are not monotonically ordered. The third finding does not match the conditional capital comparative static of Proposition 3(ii), under which lower capital raises the reset coefficient. Because open interest jointly reflects depth, price impact, trader composition, and measurement quality, we read the capacity cross-section as descriptive. The sign of capacity interactions is also sensitive to the treatment of common shocks: a continuous capacity interaction is significantly negative under complete two-way fixed effects and significantly positive under contract fixed effects alone, and the contract-level coefficients here constitute a third estimand, reinforcing the need for exogenous balance-sheet variation to identify the capital channel.⁴⁰

The diagnostic ratio $\hat{d} = (1 - A^{\text{adj}})/\kappa^{\text{eff,adj}}$ has median 7.74. In the theoretical benchmark with $m = 0$, $1 - A = d(W)\kappa^m$. Because both inputs are reduced-form projections, $\hat{d}$ serves as a scale diagnostic, and its large median value highlights a scale mismatch with a funding-only account of

³⁹Results are insensitive to the trailing-window choice: medians 0.225/0.242/0.243 for $K = 30/90/270$.

⁴⁰OA App. I reports the five-proxy fixed-effect sensitivity table. With contract and reset-time fixed effects, the interaction of $X_{i,n}$ with standardized log median open interest is -0.049 ($t = -4.90$). With contract fixed effects alone it is $+0.057$ ($t = 7.72$). The coefficients of Table 5 correlate positively with capacity across contracts (Spearman 0.214).

typical mean reversion.

7.4 Directional liquidation around basis reversals

Corollary 1 supplies a directional discriminator: in its zero-forcing benchmark, a correction that pushes the basis across zero activates margin pressure on the side opposite the pre-reversal basis (footnote 14). We compare $\text{OppLiq}_{i,n}$, liquidation notional on the side opposite $\text{sgn}(X_{i,n})$ over $[\tau_n, \tau_{n+1})$, with $\text{SameLiq}_{i,n}$ on the aligned side, and define an economic flip as a sign change of meaningful magnitude on both sides, filtering microstructure crossings.⁴¹ Two specifications implement the test,

$$\begin{aligned} \ln(1 + \text{OppLiq}_{i,n}) &= \rho^{\text{flip}} \text{Flip}_{i,n+1}^{\text{econ}} + \alpha_i + \delta_n + \varepsilon_{i,n}, \\ \ln(1 + \text{Liq}_{i,n}^s) &= \rho^s \text{HighFunding}_{i,n} + \gamma^s |X_{i,n}| + \alpha_i + \delta_n + u_{i,n}^s, \quad s \in \{\text{opp}, \text{same}\}, \end{aligned} \quad (32)$$

where $\text{HighFunding}_{i,n}$ indicates the contract's top quartile of $|f_{i,n}^8|$. The placebo contrast $\rho^{\text{opp}} - \rho^{\text{same}}$ is estimated directly, by regressing the difference $\ln(1 + \text{Liq}_{i,n}^{\text{opp}}) - \ln(1 + \text{Liq}_{i,n}^{\text{same}})$ on the same treatment, control, and fixed effects.

The data document three results (Table 6). First, intervals containing an economically meaningful sign reversal carry 0.556 more log points of opposite-side liquidation ($t = 13.0$).⁴² Because the reversal is confirmed only at the interval's closing reset while liquidation accrues during the interval, this is an interval-level association, so within-interval event ordering is unresolved. Second, the settled-funding design has cleaner temporal indexing, with funding strength observed at the reset before next-interval liquidation is measured: conditional on the dislocation's size, a top-quartile funding-strength state is associated with 0.650 more next-interval log opposite-side liquidation ($t = 15.9$). Third, the identical regression on same-side liquidation serves as a placebo: funding strength raises both sides, the same-side loading being 0.531 ($t = 13.5$), but it raises the opposite side significantly more, with a placebo-differenced loading of 0.118 ($t = 4.5$).⁴³

⁴¹The magnitude filter requires both $|X_{i,n}|$ and $|X_{i,n+1}|$ to exceed the contract's 25th percentile of $|X|$ (Appendix A).

⁴²Because zeros are common under the $\ln(1 + y)$ transformation, these coefficients are reported in log points and are not converted into percentage changes in expected liquidation notional.

⁴³The opposite-side loading is positive in every capacity quintile and declines monotonically from the thinnest to the deepest, 1.011, 0.821, 0.755, 0.542, and 0.467 (Figure 7, right panel). The placebo difference is estimated only in the pooled sample, so the asymmetry itself is not established quintile by quintile.

The opposite-minus-same contrast removes components that load both sides symmetrically, such as symmetric volatility or aggregate deleveraging, while remaining vulnerable to side-specific crowding, directional position unwinds, and differential truncation. Section 7.6 uses the same funding-state variable in a persistence specification, a different reduced-form estimand.

The liquidation feed aggregates arbitrage and directional speculative positions, and both outcomes come from the same rate-limited feed (Section 7.1). We therefore read the observed asymmetry as market-wide directional margin pressure, consistent with the flip mechanism and with crowding-based alternatives.

7.5 Sparse cap-binding states and basis persistence

Proposition 6 predicts that marginal payment feedback vanishes when the rate cap binds, and Proposition 3(iii) motivates asking whether such states also display elevated persistence, without equating cap binding with structural infeasibility. Cap binding is rare and concentrated in thin contracts (Figure 8): 563 cap-binding resets in 346 spells over the full panel, at rate 0.147% of resets in the thinnest capacity quintile against 0.011% in the deepest, a factor of about thirteen, with cap-binding resets by quintile of 116/185/133/110 against 19 in the deepest. The incidence rate is monotone in capacity, and contract-cluster bootstrap intervals separate the endpoints: $[0.099, 0.205]\%$ in the thinnest quintile against $[0.001, 0.025]\%$ in the deepest.⁴⁴

Cap-hit states are followed by greater basis persistence under both timings (Table 7): the reset loading on $|X|$ is 0.431 ($t = 4.87$) following a cap settlement at τ_n and 0.353 ($t = 4.03$) when the indicator is re-timed to the cap applying over the outcome interval. The association is concentrated in a small number of heavily treated contracts, with the influence diagnostics reported in the memo block of Table 7, and the low-capacity and triple interactions are negative, so we interpret it as a descriptive footprint of cap-limited feedback. Empirical cap spells are brief, with median one reset.⁴⁵

Around Terra/LUNA (May 2022) and FTX (November 2022), fragility, liquidation, and basis dislocations rise together, larger among exposed altcoins than among deep-market anchors, but with

⁴⁴The intervals resample contracts and therefore preserve within-contract serial dependence. Spell counts by quintile are 83/115/82/55 against 11 in the deepest.

⁴⁵OA App. I reports the empirical spell-duration table and figure.

little cap binding. The cap evidence therefore captures a narrower, contract-specific stress margin than these market-wide episodes.⁴⁶

7.6 Identification limits

We examine three dynamic comparative statics of the model: how reset persistence varies with the capital state, funding strength, and the funding interval. The available variation does not provide structural identification of these effects. The master specification (Table 8) interacts the lagged basis with a low-capital indicator and a funding-strength indicator,

$$X_{i,n+1} = \rho_1 X_{i,n} + \rho_2 X_{i,n} \times \text{LowW}_{i,n} + \rho_3 X_{i,n} \times \text{HighFunding}_{i,n} + \alpha_i + \delta_n + u_{i,n+1}, \quad (33)$$

where $\text{LowW}_{i,n}$ and $\text{HighFunding}_{i,n}$ are top-quartile indicators for predetermined fragility, a time-varying distress composite, and for settled funding strength, and time-only aggregates are absorbed by the reset-time fixed effects. Under the benchmark conditions, the model implies $\rho_2 > 0$ and $\rho_3 < 0$. The estimates do not match these benchmark signs: the fragility interaction is small, statistically insignificant, and negative (-0.009 , $t = -1.18$), whereas the funding-strength interaction is significantly positive ($+0.179$, $t = 6.9$). Neither interaction isolates its corresponding structural object: the distress composite measures latent W with error, while settled funding is mechanically tied to the recent basis.⁴⁷

Two candidate identification strategies also fail to supply usable exogenous variation in the current sample. A leave-one-out peer-liquidation instrument provides no usable first stage ($F = 1.9$). Its inputs come from the rate-limited liquidation feed of Section 7.1, which attenuates the extreme cascades most likely to generate cross-contract variation. The exchange’s own interval switches are assigned on sustained extreme funding, and the corresponding event study fails its balance and pre-trend checks.⁴⁸ The credible within-venue objects are therefore the reduced-form slopes,

⁴⁶OA App. I displays the episode diagnostics around the Terra/LUNA and FTX collapses and the event-window statistics. The cap binds at two resets on one contract across the Terra/LUNA window (XEMUSDT) and at a single reset on one contract across the FTX window (SFPUSDT). The four longest cap spells occur in mid-capacity contracts, with a single contract (TRBUSDT) accounting for three of them.

⁴⁷The two interactions correlate 0.679 and 0.554 with $X_{i,n}$. OA App. I reports the correlation matrix and the funding-strength-decile autocorrelation.

⁴⁸OA App. I reports both designs in full, including the instrument diagnostics and the event-study construction, balance tests, and pre-trend diagnostics.

clamp-region heterogeneity, and cross-sectional sorting patterns, while structural identification of the capital channel and the fixed-state interval effect remains open.

8 Conclusion and design implications

This paper develops funding as an equilibrium feedback mechanism instead of treating the posted rate as an exogenous carry. Optimizing demand and market clearing generate a capital-dependent within-interval basis law. Discrete settlement produces a stability band and flip margin. Realized arbitrage profit, margin usage, and liquidation close the capital state. Leverage, clamp, cap, and interval constraints make the reset map piecewise. The distinction among κ^m , κ^{tp1} , and κ^{eff} is essential: template parameters cannot be inserted directly into structural thresholds, and a common nominal template does not imply common effective feedback, because the reset map depends jointly on market capacity, the active branch of the rule, and settlement timing.

The empirical analysis documents three sets of reset-level facts. The state-to-settlement mapping is piecewise, with a flat center and positive slopes on either side under both observable proxies, and its measured attenuation depends on the proxy for the exchange input. Proxy-adjusted reset coefficients are positive for every contract, with the thinnest capacity quintile below the deepest. Reversal intervals carry substantially more opposite-side liquidation, strong settled funding loads more heavily on opposite- than on same-side liquidation, and cap hits concentrate in thin contracts and are followed by greater persistence. These facts are reduced form and do not separately identify latent arbitrage capital or the structural stability boundaries.

The nonlinear simulations discipline the theory from a different direction. They confirm that the local threshold geometry survives global saturation, illustrate how saturation arrests local amplification, and trace endogenous switching among constrained branches. They also show that recovery dynamics depend strongly on outside capital entry.

The model also sharpens design logic. A venue should evaluate feedback using the state-to-settlement mapping and the active piecewise branch. It should distinguish two failure modes: weak marginal payment feedback under the clamp or cap, and weak marginal arbitrage response under depleted or constrained capital. Because the reset coefficient depends on the capital state, a fixed gain cannot hold a fixed target: the coefficient drifts toward one stability boundary or the other as

capital moves, so comparable behavior across contracts calls for a gain that varies with the state. The theoretical infeasibility result says that rule tuning cannot restore local stability at all once the feasible model gain lies below the capital-dependent threshold. Identifying when a market crosses that boundary requires exogenous variation in arbitrage balance-sheet capacity, richer news measures, or cross-venue order flow.

These design implications apply when basis stabilization is part of the venue’s objective. For an exchange, the directly monitorable objects are state-to-settlement slopes, time spent in flat or capped branches, sign-flip frequency, liquidation asymmetry, and the persistence of cap states. For regulators and margin designers, the relevant question is whether a rule that appears stable in ordinary conditions becomes fragile when liquidation reduces risk-bearing capacity, and whether its cap or clamp transmits stress across contracts. For retail traders, funding is both a price-alignment device and a recurring cost of leveraged exposure, and a posted funding rate is not a guaranteed stabilizing signal. A funding rule is therefore best evaluated not by its posted formula alone but by the feedback the market actually delivers between resets.

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Figures

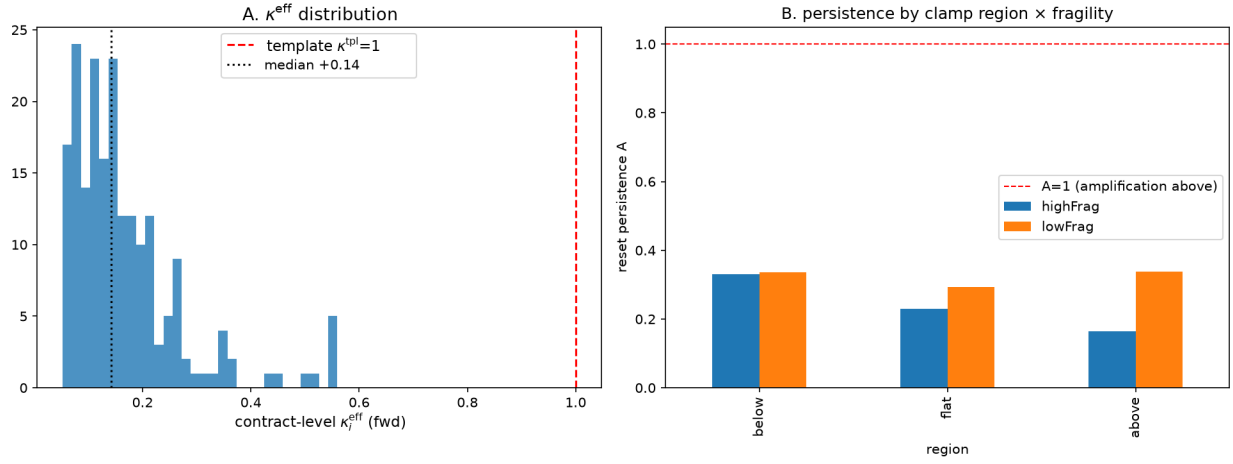


Figure 1: The reduced-form funding slope across 200 contracts versus the template anchor. Left: cross-contract distribution of κ^{eff} (forward-settled variant), with the template anchor $\kappa^{\text{tpl}} = 1$ (dashed) and the sample median 0.142 (dotted) marked. Right: reset persistence by clamp region and fragility state. Persistence stays below one throughout.

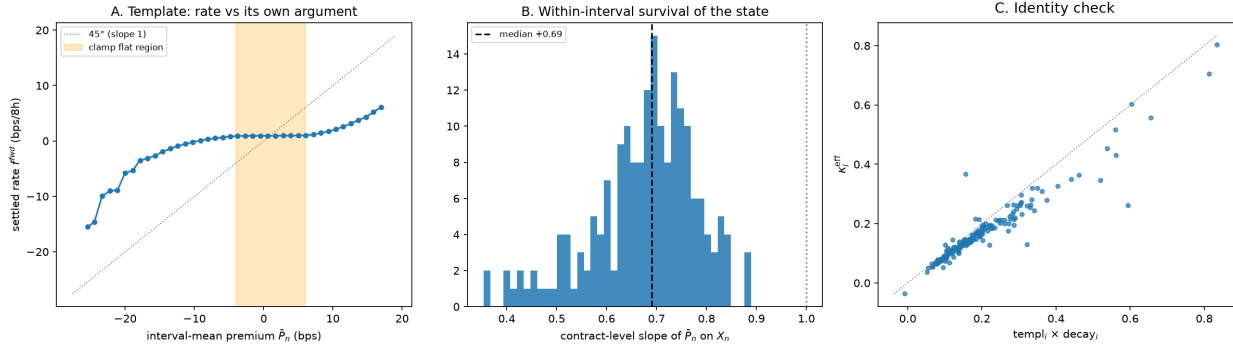


Figure 2: The piecewise rule in the data: template pass-through by clamp region. Left: settled funding rate against the interval-average premium $\bar{P}_n$ (binned means), with the clamp's flat region shaded and the 45° line marked. Middle: cross-contract distribution of the interval-retention projection (contract-level slope of $\bar{P}_n$ on X_n ; median 0.69). Right: contract-level decomposition check, plotting κ^{eff} against the product of template pass-through and retention (correlation 0.95).

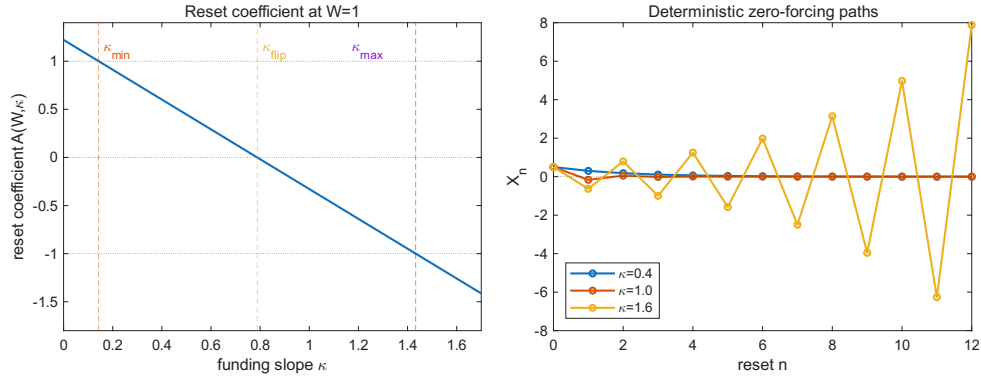


Figure 3: Theoretical stability geometry in the local snapshot benchmark. The lower boundary is the insufficient-feedback margin, the zero of A is the flip margin, and the upper boundary is the explosive-overshooting margin. Parameter values are illustrative.

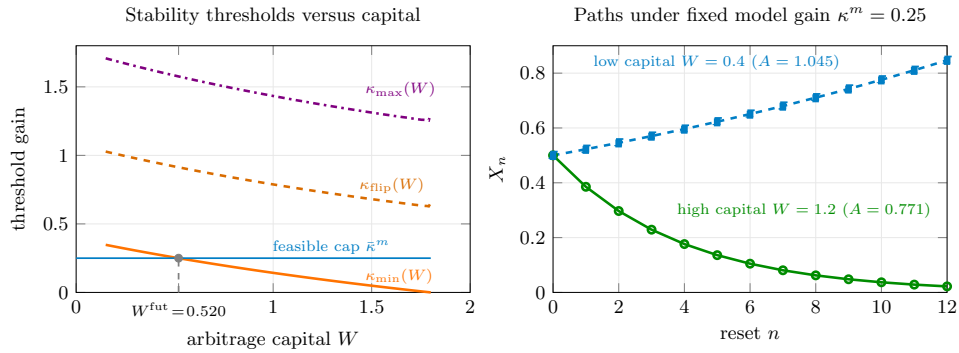


Figure 4: Capital shifts the theoretical stability region and the implied deterministic paths. The left panel shows that lower capital raises the insufficient-feedback threshold and can place a fixed feasible gain below it. The right panel applies the same fixed gain to two frozen-capital states: the high-capital path converges, whereas the low-capital path amplifies locally. Parameters are illustrative.

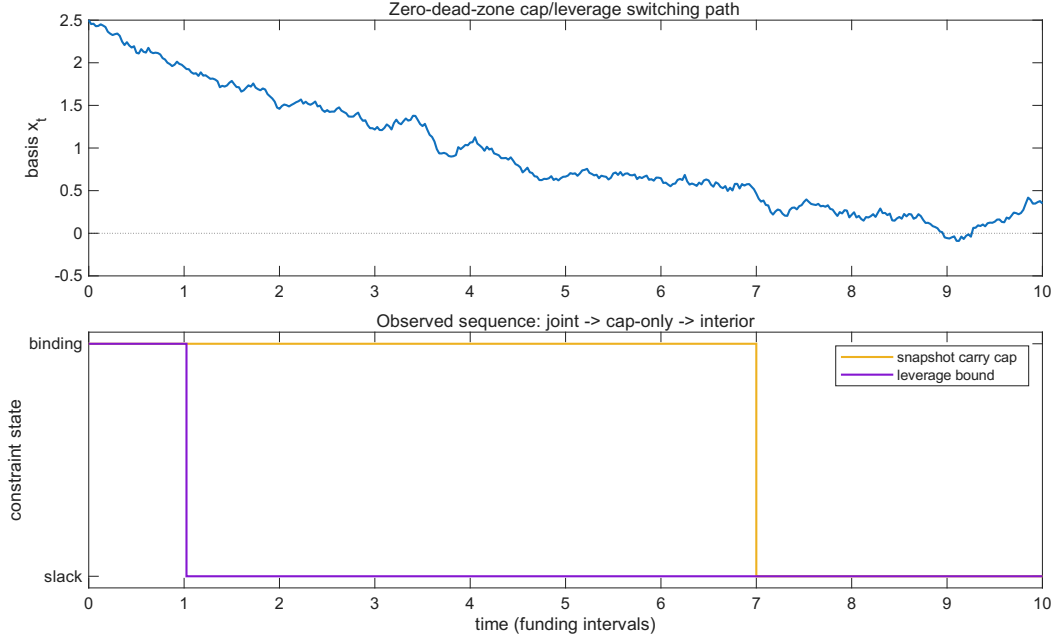


Figure 5: Illustrative switching path after a large dislocation in the zero-dead-zone snapshot benchmark. The upper panel shows the basis. The lower panel marks when the snapshot carry cap and the private leverage bound bind. Along this path the two constraints bind jointly at first, the leverage bound releases while the cap continues to bind, and the path then returns to the interior. The path uses the snapshot carry cap and leverage clipping benchmark, with the interest-centered flat region suppressed.

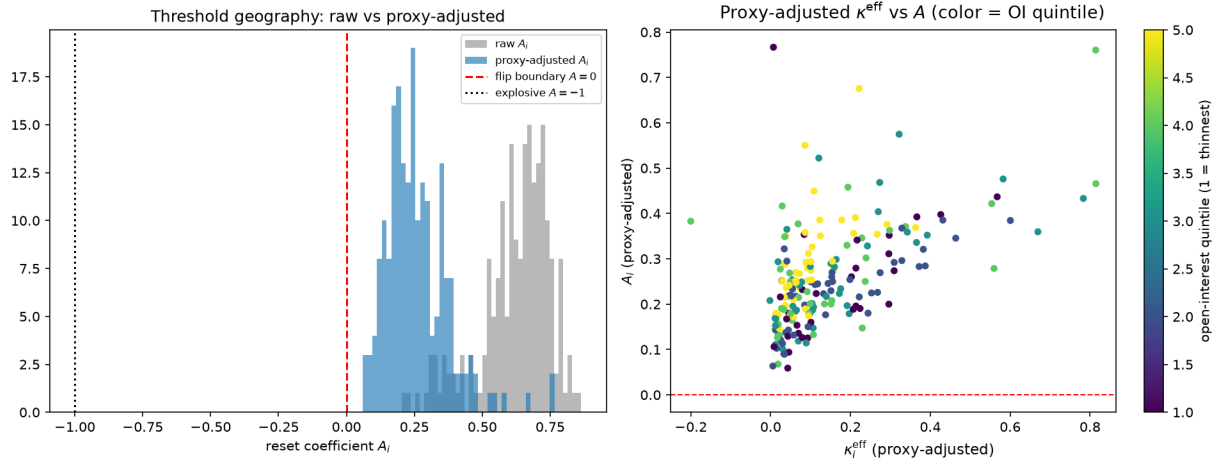


Figure 6: Raw and proxy-adjusted reset coefficients across contracts. Left: cross-contract distribution of the reset coefficient, raw (grey) and proxy-adjusted (blue), with the flip boundary $A = 0$ and the explosive boundary $A = -1$ marked. The adjustment shifts the distribution from a median of 0.65 to one of 0.24, and every contract has a positive adjusted point estimate. Right: adjusted A_i against the adjusted effective slope, colored by capacity quintile. The marked boundaries are shown for structural reference and do not classify the proxy-adjusted coefficients.

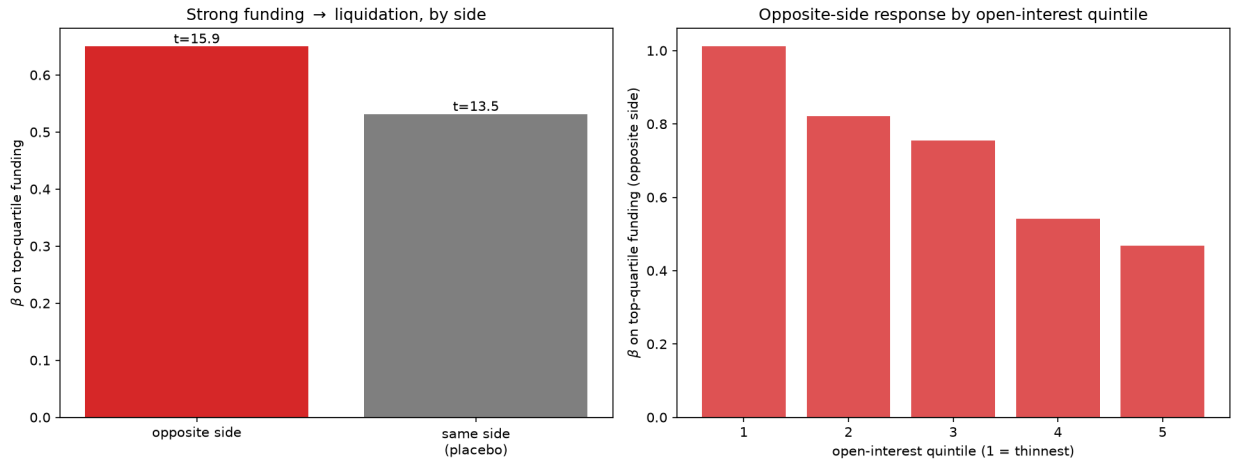


Figure 7: Sign reversals and directional liquidation. Left: loading of opposite-side and same-side (placebo) liquidation on funding strength. Stronger settled funding is associated with more liquidation on both sides, consistent with a volatility channel, and the association is significantly stronger on the opposite side. Right: the opposite-side loading by capacity quintile. Note that the right panel plots the opposite-side level, not the placebo-differenced asymmetry.

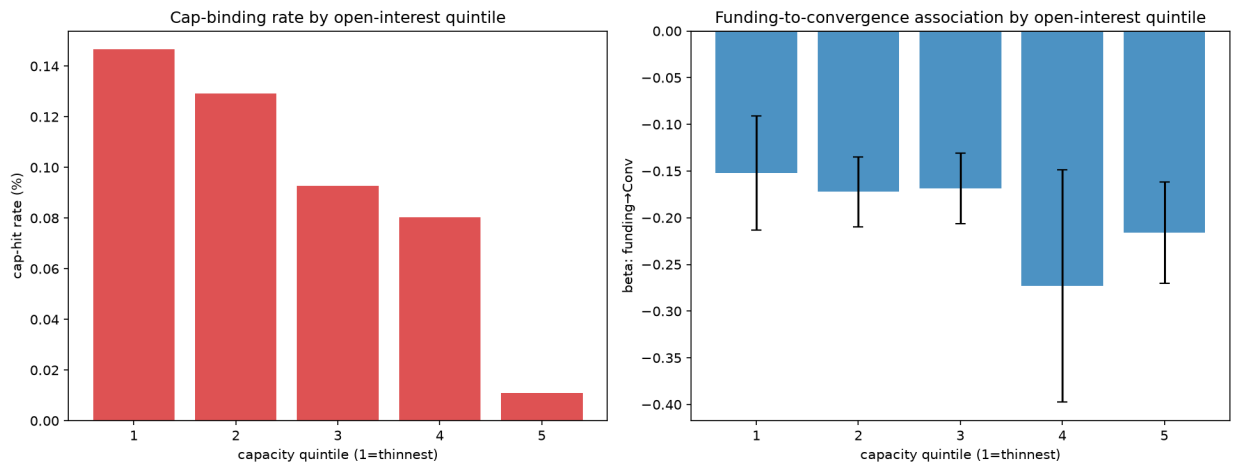


Figure 8: Cap binding and funding effectiveness across the capacity cross-section. Left: cap-binding rate by capacity quintile. The rate in the thinnest quintile is roughly thirteen times that in the deepest, and it declines monotonically from the thinnest quintile to the deepest. Right: the association between signed settled funding and next-interval convergence by quintile, with 95% intervals. The negative association in every quintile reflects the trailing-settlement state dependence discussed in Section 7.6: on a fixed template a large settled rate marks a large surviving dislocation.

Tables

Table 1: Timing and information within funding interval n .

Stage	State, information, and action
τ_n , after reset	The post-announcement basis X_n , public forcing state z_n , and current capital W_n are predetermined. The snapshot benchmark sets $c_n = \kappa^m X_n / \Delta$. The innovation ϑ_{n+1} is not public. Informed traders receive private signals and form q_n^I .
(τ_n, τ_{n+1})	Brownian shocks, decaying informed order flow, the premium path P_t , demands and positions, and marked profit accrue. The states z_n , W_n , and benchmark carry c_n remain fixed.
τ_{n+1}^-	The pre-revelation basis $\hat{X}_{n+1}$, interval profit Π_n , pre-liquidation capital $\tilde{W}_{n+1}$, and margin requirement $\mathcal{M}_{n+1}$ are determined from the completed path. The realized average $\bar{P}_n$ mechanically determines $f_{n+1}^{\text{set}} = G(\bar{P}_n)$.
τ_{n+1}	The innovation ϑ_{n+1} becomes public and the post-announcement basis X_{n+1} is observed. Funding is settled, the liquidation count L_{n+1} is drawn from the pre-revelation margin gap, and post-liquidation capital W_{n+1} becomes the state for the next interval.

Notes: The table records the model's timing convention. The hat distinguishes the pre-announcement state $\hat{X}_{n+1}$ from the post-announcement state X_{n+1} and denotes timing.

Table 2: Summary statistics, 200-contract eight-hour panel (677,405 contract-reset observations, 2020-09 to 2025-12).

Variable	N	Mean	SD	p25	Median	p75
Log basis X (bps)	677,405	-4.330	9.454	-9.145	-5.190	0
$ X $ (bps)	677,405	8.077	6.549	3.981	6.913	10.76
Convergence Conv (bps)	677,405	1.614	7.529	-2.369	0.799	4.899
Funding rate f (bps/interval)	677,405	0.3476	4.837	0.3786	1	1
Per-8h funding f_8 (bps)	677,405	0.4386	3.046	0.3786	1	1
Rolling reset coef. $\hat{A}$	673,424	0.1112	0.271	-0.07822	0.09669	0.2881
Fragility index	677,405	-0.005125	0.506	-0.2513	-0.06715	0.1531
Realized vol RV	677,405	0.02293	0.01356	0.01451	0.01979	0.02744
Liquidation intensity	677,405	0.001159	0.09357	5.97×10^{-5}	0.000283	0.000917
Taker buy/sell ratio	628,660	1.487	15.15	0.6497	0.9668	1.432
Top-trader long/short	551,464	1.544	0.7032	1.064	1.314	1.807
Cap-binding (0/1)	677,405	0.000831	0.02882	0	0	0
Economic flip (0/1)	677,405	0.02794	0.1648	0	0	0

Notes: The sampling unit is a contract-by-reset observation on the common eight-hour grid. The basis is winsorized per contract at 0.5/99.5% before differencing and reset-snapshot values beyond $\pm 5\%$ are dropped. Leads are nulled when the next reset is more than 1.5Δ away, which is why $\hat{A}$ has fewer observations than the panel. $\hat{A}$ is the 20-reset rolling within-contract AR(1) slope of X and locally demeaned, so it is not comparable to the full-sample contract-level slope. Liquidation intensity is liquidation notional over lagged open interest and is a lower bound, since the public feed is rate limited. Variable definitions are in Appendix A.

Table 3: Panel characteristics by capacity quintile.

Quintile	Contracts	Obs	Median OI	$ X $ (bps)	$\hat{A}$	Cap-hit	Flip rate
Q1 (thin)	31	79,151	\$3M	9.1	0.089	0.147%	4.28%
Q2	44	143,271	\$4M	9.0	0.096	0.129%	4.00%
Q3	40	143,331	\$6M	8.7	0.109	0.093%	2.99%
Q4	41	136,898	\$10M	7.9	0.115	0.080%	2.46%
Q5 (deep)	44	174,754	\$47M	6.5	0.132	0.011%	1.23%

Notes: Quintiles (1 = thinnest) are formed once, on the within-sample median open-interest value of each contract, so a contract keeps its capacity label throughout. Table A.1 records the underlying variable. The sort is a capacity proxy. Cap-hit is the share of resets at which $|f| \geq 0.99\bar{f}^{\text{set}}$. The flip rate uses the economic-reversal definition in Appendix A.

Table 4: Decomposing the effective slope into template pass-through and within-interval decay.

Panel A. Pooled regressions

	Dep. var. $f_{i,n}^{8,\text{fwd}}$, by clamp region of $\bar{P}$			Dep. var. $\bar{P}_{i,n}$
	(1) below	(2) inside	(3) above	(4) full sample
$\bar{P}_{i,n}$	0.856*** (22.8)	−0.009*** (−3.3)	0.253*** (15.6)	
$X_{i,n}$				0.384*** (14.8)
Contract and reset-time FE	Yes	Yes	Yes	Yes
Observations	440,468	145,396	84,521	670,385

Panel B. Contract-level statistics, 190 contracts

	Median	[IQR]
Template pass-through, non-flat	0.232	[0.17, 0.35]
Interval retention, $\bar{P}$ on X_n	0.691	[0.63, 0.75]
Effective slope κ_i^{eff}	0.143	
Product of pass-through and retention	0.161	
Ratio of κ_i^{eff} to that product	0.875	
Cross-contract correlation of the two	0.953	

Notes: $\bar{P}$ is the interval average of the 5-minute premium measure, treated as a proxy for the exchange's premium-index input. Each column of Panel A is a separate pooled regression from (30). Two-way clustered t -statistics are in parentheses below each estimate, and *, **, and *** denote significance at the 10%, 5%, and 1% levels. Columns (1)–(3) estimate κ_R^{pool} on the three disjoint subsamples defined by the clamp region of $\bar{P}$, the argument the template actually clamps, and their observation counts sum to the estimation sample. Column (4) estimates κ^{pool} on the full sample. Panel B summarizes separate contract-level regressions, one per contract, run on contracts with at least 200 non-flat observations: the first two rows give the median and interquartile range of the 190 within-contract slopes corresponding to Panel A, and the last four rows compare each contract's effective slope with the product of its own pass-through and retention, as in (28). Brackets give the interquartile range (IQR) across contracts, which measures cross-contract dispersion and is not a confidence interval for the median. The ratio row is the median of the contract-level ratios, which is a different quantity from the ratio of the two reported medians. The interval-mean proxy exists for 670,385 of the 677,405 panel observations, on 190 of the 200 contracts, which is why Panel B is estimated on a smaller universe than the median κ^{eff} of 0.142 reported in the text for all 200 contracts.

Table 5: Proxy-adjusted reset dynamics by capacity quintile.

Capacity quintile	median A^{adj}	$\mathbb{P}(A^{\text{adj}} < 0)$	median $\kappa^{\text{eff,adj}}$	median $\hat{d}$
1 (thin)	0.200 (0.57)	0%	0.093	8.32
2	0.228 (0.60)	0%	0.125	6.20
3	0.238 (0.67)	0%	0.107	6.41
4	0.233 (0.66)	0%	0.081	8.51
5 (deep)	0.275 (0.72)	0%	0.087	8.34
All	0.242 (0.65)	0%	0.096	7.74

Notes: A_i^{adj} is estimated from (31) separately for each of the 200 contracts, and $\kappa_i^{\text{eff,adj}}$ from the same regression with $f_{i,n}^{8,\text{fwd}}$ as the dependent variable. Entries are medians across the contracts in each quintile. Parentheses here contain the corresponding raw full-sample slope, not a t -statistic, and $\mathbb{P}(A^{\text{adj}} < 0)$ is the share of contracts in the quintile with a negative coefficient. The last column is the diagnostic loading $\hat{d} = (1 - A^{\text{adj}})/\kappa^{\text{eff,adj}}$. The pooled interquartile range of A^{adj} is $[0.180, 0.321]$. The raw full-sample median is 0.652. The adjusted diagnostic loading has interquartile range $[3.41, 16.77]$, with 96.0% of the 199 contracts on which it is defined at or above one. The raw comparison has interquartile range $[1.56, 4.14]$ and a share of 92.0% across all 200 contracts.

Table 6: Sign reversals and opposite-side liquidation.

	(1) Opposite side	(2) Same side (placebo)	(3) Difference (1)–(2)
<i>Panel A. Economic reversal; dependent variable $\ln(1 + \text{OppLiq})$</i>			
Economic flip	0.556*** (13.01)	—	—
<i>Panel B. Strong-funding treatment; dependent variable $\ln(1 + \text{Liq}_{i,n}^s)$ on the indicated side</i>			
Top-quartile $ f_{i,n}^8 $	0.650*** (15.93)	0.531*** (13.45)	0.118*** (4.49)
Control for $ X_{i,n} $	Yes	Yes	Yes
Contract and reset-time FE	Yes	Yes	Yes
Observations, each regression		677,405	

Notes: The two panels report ρ^{flip} and ρ^s from (32), estimated on the 200-contract panel. Two-way clustered t -statistics are in parentheses below each estimate, and *, **, and *** denote significance at the 10%, 5%, and 1% levels. OppLiq and SameLiq are liquidation notional over $[\tau_n, \tau_{n+1})$ on the side opposite to, and aligned with, $\text{sgn}(X_{i,n})$, and both panels measure the outcome over that window and enter as $\ln(1 + \text{notional})$. A liquidation order recorded as a sell is a liquidated long position and one recorded as a buy is a liquidated short position. For $X_{i,n} > 0$ the opposite side is therefore the long side, measured from sell-recorded orders, and symmetrically for $X_{i,n} < 0$. An economic flip requires a sign change with $|X_{i,n}|$ and $|X_{i,n+1}|$ both above the contract's 25th percentile of $|X|$. The reversal specification of Panel A is estimated only for the opposite side, so its placebo and difference cells are empty by construction. Column (3) of Panel B is a separate regression of the difference $\ln(1 + \text{OppLiq}) - \ln(1 + \text{SameLiq})$ on the same treatment, control, and fixed effects. The Panel B treatment is an indicator for the contract's top quartile of $|f_{i,n}^8|$, so all coefficients are in log points of the outcome. Because zeros are common under the $\ln(1 + y)$ transformation, they are not converted into percentage changes in liquidation notional.

Table 7: Cap binding and basis persistence.

	Dependent variable: $ X _{i,n+1}$	
	Coefficient	Benchmark sign
$ X _{i,n}$	0.2628*** (12.42)	—
$ X _{i,n} \times \text{CapHit}_{i,n}$	0.4661*** (10.19)	> 0
$ X _{i,n} \times \text{LowCap}_i$	−0.0149 (−0.46)	> 0
$ X _{i,n} \times \text{CapHit}_{i,n} \times \text{LowCap}_i$	−0.1653*** (−2.75)	> 0
Contract and reset-time FE	Yes	
<i>Memo: $X _{i,n} \times \text{CapHit}_{i,n}$ estimated without the capacity interactions</i>		
Coefficient (t)	0.431 (4.87)	
Forward-timing variant, $\text{CapHit}_{i,n}^{\text{fwd}}(t)$	0.353 (4.03)	
Range dropping one of the 15 most treated contracts	[0.224, 0.454]	
Coefficient dropping the ten longest spells (t)	0.153 (3.25)	
Share of cap-binding resets in the top five contracts	21.5%	

Notes: The regression is $|X|_{n+1}$ on $|X|_n$ interacted with cap-binding and thin-capacity indicators and their product, with contract and reset-time fixed effects. Two-way clustered t -statistics are in parentheses below each estimate, and *, **, and *** denote significance at the 10%, 5%, and 1% levels. CapHit indicates $|f_{i,n}| \geq 0.99 \bar{f}^{\text{set}}$ with $\bar{f}^{\text{set}} = 0.75\%$, where $f_{i,n}$ is the rate settled at τ_n . Specifications using it therefore describe persistence following a cap settlement, one reset earlier than the cap branch of the theory. The cap is taken as the standard template value 0.75% for all contracts and dates. Contract-specific or date-specific cap adjustments, if any, are not reconstructed. The forward variant in the memo block re-times the indicator to $\text{CapHit}_{i,n}^{\text{fwd}} = \text{CapHit}_{i,n+1}$ on consecutive resets, aligning it with the funding applying over the outcome interval, and loses one reset per contract (677,205 observations against 677,405). LowCap is the thinnest capacity group. The benchmark-sign column applies the derivative conditions of Proposition 3(ii) with a nonnegative reset coefficient on the active branch, reading cap binding as a low-capital footprint. The memo block reports the cap interaction estimated without the two capacity terms, together with influence diagnostics for that estimate. The deletion entries carry no separate significance test.

Table 8: The master specification of the lagged basis and its state interactions.

Dependent variable: $X_{i,n+1}$		
	Coefficient	Benchmark sign
$X_{i,n}$ (ρ_1)	0.2436*** (11.11)	—
$X_{i,n} \times \text{LowW}_{i,n}$ (ρ_2)	−0.0086 (−1.18)	> 0
$X_{i,n} \times \text{HighFunding}_{i,n}$ (ρ_3)	0.1792*** (6.91)	< 0
Contract and reset-time FE	Yes	
Observations	677,405	

Notes: The regression is (33), estimated with contract and reset-time fixed effects. Two-way clustered t -statistics are in parentheses below each estimate, and *, **, and *** denote significance at the 10%, 5%, and 1% levels. Time-only aggregates are absorbed by the reset-time fixed effects. LowW is top-quartile predetermined fragility, a time-varying distress composite, not the cross-sectional capacity sort used elsewhere, and HighFunding is top-quartile $|f_{i,n}^8|$. The benchmark-sign column states the sign implied by the model's comparative statics under their sufficient conditions. The capital sign in particular is conditional.

A Variable definitions

Table A.1: Variable definitions.

Variable	Definition
$X_{i,n}$	reset-time log basis, $\log F_{i,n} - \log S_{i,n}$, from 5-minute mid prices at the reset; lead $X_{i,n+1}$ defined for consecutive resets only
$\text{Conv}_{i,n}$	signed convergence, $-(X_{i,n+1} - X_{i,n}) \cdot \text{sgn}(X_{i,n})$
$f_{i,n}, f_{i,n}^8$	realized funding settled at reset n ; per-8h normalization $f_{i,n}^8 = f_{i,n} \cdot (8/\Delta)$
$f_{i,n}^{8,\text{fwd}}$	forward-settled per-8h rate computed from interval- n observations and settled at τ_{n+1} ; it is the outcome used for κ^{eff} and is not predetermined at τ_n
$\bar{P}_{i,n}$	interval average of the 5-minute perpetual-minus-spot basis; it is a proxy for the exchange input P_t . A second proxy, the interval average of 5-minute premium-index klines, is used in the measurement comparison of Section 7.2
$\text{CapHit}_{i,n}$	$\mathbf{1}\{ f_{i,n} \geq 0.99 \bar{f}^{\text{set}}\}$ with $\bar{f}^{\text{set}} = 0.75\%$, the standard template cap; contract-date cap adjustments, if any, are not reconstructed
$\text{Frag}_{i,n}$	$\frac{1}{4}[z(\text{LIQINT}_{i,n-1}) + z(\text{OIDD}_{i,n}) + z(\text{Crowd}_{i,n}) + z(\text{RV}_{i,n})]$: liquidation notional over lagged open interest, open-interest drawdown, $ \log(\text{long/short}) $, and interval realized volatility
$\text{LowW}_{i,n}$	$\mathbf{1}\{\text{predetermined fragility in the contract's top quartile}\}$
$\text{HighFunding}_{i,n}$	$\mathbf{1}\{ f_{i,n}^8 \text{ in the contract's top quartile}\}$
$\hat{A}_{i,n}$	20-reset rolling within-contract AR(1) slope of X
A_i^{adj}	proxy-adjusted reset coefficient: contract-level slope of X_{n+1} on X_n controlling for the trailing 90-reset own mean and the leave-one-out cross-sectional mean basis; $\kappa_i^{\text{eff,adj}}$ defined with the same controls
$\text{Flip}_{i,n+1}^{\text{econ}}$	$\mathbf{1}\{X_{i,n}X_{i,n+1} < 0, X_{i,n} , X_{i,n+1} \text{ above the contract's 25th percentile of } X \}$
$\text{OppLiq}_{i,n}, \text{SameLiq}_{i,n}$	liquidation notional over $[\tau_n, \tau_{n+1})$ on the side opposite to (aligned with) $\text{sgn}(X_{i,n})$; for $X_{i,n} > 0$ the opposite side is the long side and the aligned side the short side, symmetrically for $X_{i,n} < 0$
cap_rank_i	within-sample quintile of median open-interest value (1 = thinnest)

Notes: Subscripts denote contract i and reset n , and $z(\cdot)$ denotes within-contract standardization. All variables are built at the contract-reset frequency on the common eight-hour grid. Prices are 5-minute mid quotes at the reset. Liquidation series come from the rate-limited public feed and therefore understate notional, most severely in extreme states. The empirical objects are reduced-form counterparts of the model objects: $\bar{P}_{i,n}$ proxies the exchange input P_t , A_i^{adj} is a proxy-adjusted projection coefficient, and Frag and cap_rank proxy the latent capacity and distress states.

B Proofs and derivations for the main theoretical results

This print appendix contains the derivations and proofs on which the empirical interpretation depends. The main text displays the core nonlinear simulations. The OA records their implementation and sensitivities, together with additional aggregation and analytical extensions not needed for the main empirical interpretation.

B.1 Equilibrium construction and reset equations

Proof of Proposition 1. The retail problem (3) is strictly concave with unique policy $q_{i,t} = \eta_i(s_{i,t} - c_n)$. Under the capacity-weighted cancellation $\int_0^1 \eta_i \xi_{i,t} di = 0$, aggregation and local linearization yield

$$\begin{aligned} q_t^M &= \eta_M [\mu_M(x_t) + \gamma z_n - c_n] = \eta_M (\beta_M x_t + \gamma z_n - c_n) + o(x_t), \\ y_t &= \eta_A W_n [\mu_A(x_t) + c_n] = \eta_A W_n (\beta_A x_t + c_n) + o(x_t), \quad q_t^A = -y_t. \end{aligned} \quad (\text{B.1})$$

The second policy follows from the strictly concave arbitrage problem (5) on its slack branch.

Signed customer net notionals are $D_t^F = q_t^M + q_t^A + q_t^{I,F}$ in the perpetual and $D_t^S = q_t^{I,S} - q_t^A$ in spot, where the informed venue flows are $q_t^{I,F} = \theta_I e^{-\rho_I(t-\tau_n)} q_n^I$ and $q_t^{I,S} = (1 - \theta_I) e^{-\rho_I(t-\tau_n)} q_n^I$. Intermediary inventories clear as $\mathcal{H}_t^j = -D_t^j$. The two competitive intermediary problems (6) are strictly concave. Substituting the clearing conditions into the first-order conditions gives

$$\varpi_t^F = \frac{D_t^F}{\lambda_F} - \phi_F x_t - c_n, \quad \varpi_t^S = \frac{D_t^S}{\lambda_S} + \phi_S x_t.$$

The common process v_t is an otherwise unrestricted common log-valuation semimartingale. The processes $\tilde{B}_t^F$ and $\tilde{B}_t^S$ are venue-level Brownian shocks with loadings $\sigma_F, \sigma_S \geq 0$. Define the standard basis Brownian motion and its volatility by $\sigma_x dB_t \equiv \sigma_F d\tilde{B}_t^F - \sigma_S d\tilde{B}_t^S$, with $\sigma_x > 0$ incorporating their instantaneous covariance. The associated venue-price equations are

$$\begin{aligned} d \log F_t &= dv_t + (\alpha_F D_t^F - \phi_F x_t - c_n) dt + \sigma_F d\tilde{B}_t^F, \\ d \log S_t &= dv_t + (\alpha_S D_t^S + \phi_S x_t) dt + \sigma_S d\tilde{B}_t^S. \end{aligned} \quad (\text{B.2})$$

Insert (B.1) and the informed schedules into (B.2), then subtract the spot equation from the perpetual

one. The common dv_t cancels. The coefficient on x_t is

$$-\phi_F - \phi_S + \alpha_F \eta_M \beta_M - (\alpha_F + \alpha_S) \eta_A \beta_A W_n = a - bW_n,$$

while the coefficient on c_n is

$$-1 - \alpha_F \eta_M - (\alpha_F + \alpha_S) \eta_A W_n = -d(W_n).$$

The public and informed coefficients are ψ and $-\chi_I e^{-\rho_I(t-\tau_n)}$, respectively. This proves (7) and (8), in which z_n and q_n^I enter the drift additively. Given the predetermined state and a shock path, the resulting local linear stochastic differential equation has a unique strong solution. $\square$

To derive the reset map, write interval time as $s = t - \tau_n$ and define $B_{n,r} \equiv B_{\tau_n+r} - B_{\tau_n}$ for $r \in [0, \Delta]$. Variation of constants applied to (7) gives

$$\begin{aligned} x_{\tau_n+s} &= e^{m(W_n)s} X_n + [\psi z_n - d(W_n) c_n] \int_0^s e^{m(W_n)(s-r)} dr \\ &\quad - \chi_I q_n^I \int_0^s e^{m(W_n)(s-r)} e^{-\rho_I r} dr + \sigma_x \int_0^s e^{m(W_n)(s-r)} dB_{n,r}. \end{aligned} \quad (\text{B.3})$$

At $s = \Delta$, the informed-flow kernel is

$$\mathcal{J}(m, \rho, \Delta) = \int_0^\Delta e^{m(\Delta-s)-\rho s} ds = \begin{cases} \frac{e^{m\Delta} - e^{-\rho\Delta}}{m + \rho}, & m + \rho \neq 0, \\ \Delta e^{m\Delta}, & m + \rho = 0. \end{cases} \quad (\text{B.4})$$

so the informed loading is $B_I(W) \equiv \chi_I \mathcal{J}(m(W), \rho_I, \Delta)$. The integrated Brownian innovation is

$$\varepsilon_{n+1}^B = \sigma_x \int_0^\Delta e^{m(W_n)(\Delta-s)} dB_{n,s},$$

and the pre-announcement innovation in (9) is $\widehat{\varepsilon}_{n+1} = -B_I(W_n) q_n^I + \varepsilon_{n+1}^B$. Substitution of (2) into (B.3) yields (9) and (10). Adding the announcement jump $J_z \vartheta_{n+1}$, with $J_z \in \mathbb{R}$ the reset-auction basis loading per unit innovation, and using $q_n^I = \eta_I \vartheta_{n+1}$ gives (11) with net loading $\Gamma_I(W) = J_z - \eta_I B_I(W)$. The limits at $m(W) = 0$ follow from $(e^{m\Delta} - 1)/m \rightarrow \Delta$.

B.2 Stability and measurement

Proof of Proposition 2. For $m(W) \neq 0$,

$$\frac{\partial A(W, \kappa^m)}{\partial \kappa^m} = -\frac{d(W)}{\Delta} \frac{e^{m(W)\Delta} - 1}{m(W)} < 0,$$

and the continuous limit is $-d(W) < 0$. Thus A is strictly decreasing and $A(W, 0) = e^{m(W)\Delta}$.

When $m(W) > 0$, $A(W, 0) > 1$. Because A is affine in κ^m , the roots of $A = 1$, $A = 0$, and $A = -1$ are precisely $\kappa_{\min}$, κ_{flip} , and $\kappa_{\max}$ in (12). Their ordering follows from

$$\frac{\kappa_{\text{flip}}}{\kappa_{\min}} = \frac{e^{m\Delta}}{e^{m\Delta} - 1} > 1, \quad \frac{\kappa_{\max}}{\kappa_{\text{flip}}} = \frac{e^{m\Delta} + 1}{e^{m\Delta}} > 1.$$

Iteration of the zero-forcing recursion gives $X_{n+j} = A^j X_n$, establishing the sign and magnitude regions.

When $m(W) < 0$, $A(W, 0) \in (0, 1)$, so the lower endpoint is zero and stability fails only at the root of $A = -1$. At $m(W) = 0$, $A = 1 - d(W)\kappa^m$, giving the continuous boundaries $\kappa_{\text{flip}} = 1/d(W)$ and $\kappa_{\max} = 2/d(W)$. $\square$

Derivation of (29). Write $\Sigma_X = \text{Var}(X_n)$ and $C_{Xz} = \text{Cov}(X_n, z_n)$. Taking the covariance of (11) with X_n and using $\text{Cov}(\varepsilon_{n+1}, X_n) = 0$, which follows from the conditional mean of the reset innovation given the predetermined state, gives $\text{Cov}(X_{n+1}, X_n) = A\Sigma_X + \Psi C_{Xz}$. Division by $\Sigma_X > 0$ gives (29). No stationarity is used.

If the system (4)–(11) is in addition covariance stationary with $|A| < 1$ and $|\phi_z| < 1$, then $\Sigma_z = \sigma_{\vartheta}^2 / (1 - \phi_z^2)$, and taking the covariance of (11) at $n+1$ with (4) at $n+1$ yields

$$C_{Xz} = A\phi_z C_{Xz} + \phi_z \Psi \Sigma_z + \Gamma_I \sigma_{\vartheta}^2,$$

because the Brownian innovation is independent and $\text{Cov}(\Gamma_I \vartheta_{n+1}, \vartheta_{n+1}) = \Gamma_I \sigma_{\vartheta}^2$. Solving gives

$$C_{Xz} = \frac{\phi_z \Psi(W) \Sigma_z + \Gamma_I(W) \sigma_{\vartheta}^2}{1 - \phi_z A(W, \kappa^m)}, \quad \Sigma_z = \frac{\sigma_{\vartheta}^2}{1 - \phi_z^2}. \quad (\text{B.5})$$

Under stationarity $\text{Var}(X_{n+1}) = \text{Var}(X_n)$, so β^{raw} also equals the one-reset correlation.

If z_n is observed and the conditional-mean restriction holds, the Frisch–Waugh–Lovell residual of (11) on z_n has coefficient A on residualized X_n . Replacing z_n by proxies preserves equality only under the stated spanning and orthogonality conditions. $\square$

Derivation of (27). With additive forcing suppressed, the conditional mean path on the active branch is

$$\mathbb{E}_n[x_{\tau_n+s}] = e^{m(W)s} X_n - d(W) c_n \frac{e^{m(W)s} - 1}{m(W)}.$$

Averaging over $s \in [0, \Delta]$ and using $P_t = \lambda_P x_t$ gives

$$\mathbb{E}_n[\bar{P}_n] = \lambda_P h(u) X_n - \lambda_P d(W) c_n \Delta g(u).$$

Because $c_n \Delta = \kappa^{\text{tpl}} \mathbb{E}_n[\bar{P}_n]$, this is a conditional-mean fixed point in $\mathbb{E}_n[\bar{P}_n]$. Solving it gives

$$\mathbb{E}_n[\bar{P}_n] = \frac{\lambda_P h(u)}{1 + d(W) \kappa^{\text{tpl}} \lambda_P g(u)} X_n,$$

and $\mathbb{E}_n[f_{n+1}^{\text{act}}] = \kappa^{\text{tpl}} \mathbb{E}_n[\bar{P}_n]$ gives (27). Evaluating the mean path at $s = \Delta$ and substituting the same fixed point gives the reset-coefficient counterpart recorded in OA App. E. $\square$

B.3 Capital fragility and directional liquidation

Proof of Proposition 3. On the stressed region,

$$\kappa_{\min}(W) = \Delta \frac{a - bW}{1 + d_0 + d_1 W}.$$

Differentiating gives

$$\kappa'_{\min}(W) = -\Delta \frac{b(1 + d_0) + ad_1}{(1 + d_0 + d_1 W)^2} < 0,$$

which is (16). Evaluating the same function at $(1 - \delta)W$ and W and taking a common denominator gives the exact capital-loss jump:

$$\kappa_{\min}((1 - \delta)W) - \kappa_{\min}(W) = \frac{\Delta \delta W [ad_1 + b(1 + d_0)]}{(1 + d_0 + d_1 W)[1 + d_0 + d_1(1 - \delta)W]} > 0.$$

This proves part (i).

For part (ii), write (10) as $A = e^u - d(W)\kappa^m h(u)$, where $u = (a - bW)\Delta$ and $h(u) = (e^u - 1)/u$. Since $du/dW = -b\Delta$ and $d'(W) = d_1$,

$$\frac{\partial A}{\partial W} = -b\Delta e^u - \kappa^m [d_1 h(u) - b\Delta d(W)h'(u)],$$

which is (17). If $d_1 h(u) - b\Delta d(W)h'(u) \geq 0$, the first term is strictly negative and the second is nonpositive. If the bracket is negative, moving its positive contribution to the other side shows that $\partial A/\partial W < 0$ if and only if

$$\kappa^m < \frac{b\Delta e^u}{b\Delta d(W)h'(u) - d_1 h(u)}.$$

This establishes the conditional sign statement.

For part (iii), Proposition 2 implies that a feasible gain stabilizes the stressed interior branch precisely when $[0, \bar{\kappa}^m]$ intersects the open interval $(\kappa_{\min}(W), \kappa_{\max}(W))$. Because $\kappa_{\max}(W) > \kappa_{\min}(W)$, that intersection is empty if and only if $\bar{\kappa}^m \leq \kappa_{\min}(W)$. Solving $\bar{\kappa}^m = \Delta(a - bW)/(1 + d_0 + d_1 W)$ gives

$$W = \frac{\Delta a - \bar{\kappa}^m(1 + d_0)}{\bar{\kappa}^m d_1 + \Delta b} = W^{\text{fut}},$$

which is (18). Its numerator is positive exactly when $\bar{\kappa}^m < \Delta a/(1 + d_0)$. Strict monotonicity of $\kappa_{\min}(W)$ then makes this boundary unique and implies $\bar{\kappa}^m \leq \kappa_{\min}(W)$ exactly for $W \leq W^{\text{fut}}$. $\square$

Proof of Corollary 1. In the homogeneous no-innovation benchmark, take $X_n \neq 0$. Then

$$\hat{X}_{n+1} = A(W_n, \kappa^m)X_n.$$

Proposition 2 gives $A < 0$ above the flip threshold, so the deterministic reset state has the sign opposite to X_n . If $X_n > 0$, then $[\hat{X}_{n+1}]_- = -AX_n > 0$, and the newly activated term inside (15) is $\Delta\lambda_x^- A^2 X_n^2$. If $X_n < 0$, the symmetric term is $\Delta\lambda_x^+ A^2 X_n^2$. At the flip threshold $A = 0$. Beyond it, $A < 0$ and $\partial A/\partial \kappa^m < 0$, so

$$\frac{\partial A^2}{\partial \kappa^m} = 2A \frac{\partial A}{\partial \kappa^m} > 0.$$

Hence the opposite-side component is weakly increasing in κ^m and strictly increasing whenever the relevant $\lambda_x^\mp > 0$. At or below the flip threshold, $A \geq 0$, and the deterministic component does not cross to the opposite side. This proves the corollary. $\square$

Proof of Proposition 4. Under (19), $0 < 1 - \delta < 1$ and $\Lambda_n > 0$ imply $W_{n+1} < W_n$. Proposition 3(i) then implies $\kappa_{\min}(W_{n+1}) > \kappa_{\min}(W_n)$, proving part (i).

For part (ii), when $W_0 > W^{\text{fut}}$,

$$W_1 \leq W^{\text{fut}} \iff (1 - \delta)^{\Lambda_0} \leq \frac{W^{\text{fut}}}{W_0}.$$

Both ratios lie in $(0, 1)$. Taking logarithms and dividing by the negative number $\log(1 - \delta)$ reverses the inequality:

$$\Lambda_0 \geq \frac{\log(W^{\text{fut}}/W_0)}{\log(1 - \delta)}.$$

This is the condition stated in the proposition. The integer-count extension is reserved for OA App. C. $\square$

Lemma 1 (Inward tail drift under saturated demand). *Fix an interval state (W_n, z_n, q_n^I) with $W_n > 0$. Suppose μ_M is bounded, arbitrage exposure satisfies $|y_t| \leq \bar{\ell}W_n$, and $|c_n| \leq \bar{c}$ for a finite per-unit-time snapshot carry cap $\bar{c} > 0$. Then there is a finite, state-dependent radius R_n such that the conditional global-basis drift satisfies $xb_x(x) < 0$ whenever $|x| > R_n$.*

Proof of Lemma 1. Return to the nonlinear demand schedules before localizing μ_M and μ_A . Conditional on the fixed interval state, every contribution to the basis drift other than intermediary resiliency is bounded. Retail demand is bounded because μ_M saturates. Arbitrage exposure is bounded by $\bar{\ell}W_n$. Funding is bounded by $\bar{c}$. Public and informed order flow are fixed at finite (z_n, q_n^I) . Hence the conditional drift can be written

$$b_x(x) = -\phi_x x + r_n(x), \quad |r_n(x)| \leq C_n < \infty.$$

Because $\phi_x > 0$,

$$xb_x(x) \leq -\phi_x x^2 + C_n |x| < 0 \quad \text{whenever} \quad |x| > \frac{C_n}{\phi_x}.$$

Taking $R_n = C_n/\phi_x$ proves the stated inward-drift inequality. The argument is conditional and does not by itself establish positive recurrence of the joint basis–capital process. $\square$

B.4 Binding and exchange-rule branches

Proof of Proposition 5. Condition on a fixed-sign leverage corner remaining valid for the entire interval. The resulting affine recursion is exact on that branch. The Kuhn–Tucker conditions for (5) project the unconstrained target $W_n \eta_A [\mu_A(x_t) + c_n]$ onto $[-\bar{\ell} W_n, \bar{\ell} W_n]$, yielding (20). On a fixed-sign corner, substitute $y_t = \zeta_n \bar{\ell} W_n$ into the demand and price equations. The marginal terms $-b W_n x_t$ and $-d_1 W_n c_n$ disappear, leaving

$$dx_t = [ax_t + \psi z_n - (1 + d_0)c_n - \zeta_n \alpha_A \bar{\ell} W_n - \chi_I e^{-\rho_I(t-\tau_n)} q_n^I] dt + \sigma_x dB_t.$$

Integrating this equation over the interval, substituting $c_n = \kappa^m X_n / \Delta$, and adding the reset announcement produces the full affine recursion

$$\begin{aligned} X_{n+1} &= A^{\text{bind}}(\kappa^m) X_n + \Psi^{\text{bind}} z_n - \zeta_n C^{\text{bind}}(W_n) + \epsilon_{n+1}^{\text{bind}}, \\ \Psi^{\text{bind}} &= \psi \frac{e^{a\Delta} - 1}{a}, \quad C^{\text{bind}}(W) = \alpha_A \bar{\ell} W \frac{e^{a\Delta} - 1}{a}, \end{aligned} \tag{B.6}$$

with continuous limits at $a = 0$. The informed-flow, Brownian, and announcement components are collected in $\epsilon_{n+1}^{\text{bind}}$. The coefficient on X_n is exactly $A^{\text{bind}}(\kappa^m)$ in (21). Since $(e^{a\Delta} - 1)/a > 0$, the corner inventory shifts the reset state in the economically correct direction and $C^{\text{bind}}(W) > 0$.

When $a > 0$, $A^{\text{bind}}(0) = e^{a\Delta} > 1$ and A^{bind} is strictly decreasing in κ^m . Solving $A^{\text{bind}} = 1$ and $A^{\text{bind}} = -1$ gives the two boundaries in (22). The branch is therefore stable exactly between them. Finally, subtracting the interior lower threshold from the binding lower threshold gives

$$\begin{aligned} \kappa_{\min}^{\text{bind}} - \kappa_{\min}(W) &= \frac{a\Delta}{1 + d_0} - \frac{(a - bW)\Delta}{1 + d_0 + d_1 W} \\ &= \frac{\Delta W [ad_1 + b(1 + d_0)]}{(1 + d_0)(1 + d_0 + d_1 W)} > 0. \end{aligned}$$

Thus the binding branch can fail the lower stability condition even when the interior branch at the same capital level remains stabilizable. $\square$

Proof of Proposition 6. This proof concerns the normalized snapshot map (24), not the literal

institutional map G . For any carry fixed within an interval, (B.3) implies

$$X_{n+1} = e^{m(W_n)\Delta} X_n + \Psi(W_n) z_n - H(W_n) \Delta c_n + \varepsilon_{n+1},$$

where $H(W) = d(W)[e^{m(W)\Delta} - 1]/[\Delta m(W)]$ for $m(W) \neq 0$ and $H(W) = d(W)$ at the continuous limit. Substituting $\Delta c_n = g_{x_{dz}, \bar{c}}(X_n)$ gives the first line of (25). To record the level effects used in that substitution, write $m = m(W_n)$, $H = H(W_n)$, and $\Psi = \Psi(W_n)$. The three branches are

$$X_{n+1} = \begin{cases} e^{m\Delta} X_n + \Psi z_n + \varepsilon_{n+1}, & |X_n| \leq x_{dz}, \\ A(W_n, \kappa^m) X_n + H \kappa^m x_{dz} \operatorname{sgn}(X_n) + \Psi z_n + \varepsilon_{n+1}, & x_{dz} < |X_n| \leq x_{dz} + \Delta \bar{c} / \kappa^m, \\ e^{m\Delta} X_n - H \Delta \bar{c} \operatorname{sgn}(X_n) + \Psi z_n + \varepsilon_{n+1}, & |X_n| > x_{dz} + \Delta \bar{c} / \kappa^m. \end{cases} \quad (\text{B.7})$$

In the flat and capped branches, the payment is locally constant in X_n , so differentiation gives $e^{m\Delta}$. In the intermediate branch, $g = \kappa^m X_n - \kappa^m x_{dz} \operatorname{sgn}(X_n)$ and $e^{m\Delta} - H \kappa^m = A(W_n, \kappa^m)$, giving the middle derivative in (25). The derivative is undefined at the four kink points, as stated in the proposition. $\square$

Proof of Corollary 2. Under zero forcing, $|X_{n+j}| = |A(W, \kappa^m; \Delta)|^j |X_n|$. Solving $|A|^j = 1/2$ gives the step half-life. Multiplying by Δ gives calendar time. Replacing A by a fixed target $\bar{A}$ proves part (ii).

For part (iii), write $u = m(W)\Delta$, $q = d(W)\kappa^m$, and $A = e^u - qh(u)$. Holding W and the per-interval gain fixed,

$$\frac{\partial A}{\partial \Delta} = m(W)[e^u - qh'(u)].$$

For $u > 0$, $h'(u) > 0$ and the upper stable boundary in q -space is $q_{\max} = (e^u + 1)/h(u)$. Direct simplification gives

$$e^u - q_{\max} h'(u) = \frac{e^{2u} - 1 - 2ue^u}{u(e^u - 1)} = \frac{2e^u(\sinh u - u)}{u(e^u - 1)} > 0.$$

Hence $e^u - qh'(u) > 0$ for every $q < q_{\max}$ and $\partial A / \partial \Delta > 0$ throughout the stable band when $m(W) > 0$. If $u < 0$, the derivative is negative at $q = 0$, whereas the last displayed bracket is negative at $q = q_{\max}$ and therefore positive after multiplication by $m(W) < 0$. Values just inside the upper

boundary retain that sign. Thus both signs are possible when $m(W) < 0$. $\square$

B.5 Projection from institutional inputs to empirical slopes

Derivation of (28). Take the covariance of the settled-rate projection $f_{i,n}^{8,\text{fwd}} = \alpha_i^f + \kappa_i^{\text{avg}} \bar{P}_{i,n} + e_{i,n}^f$ with $X_{i,n}$ and divide by $\text{Var}_i(X_{i,n}) > 0$:

$$\frac{\text{Cov}_i(f_{i,n}^{8,\text{fwd}}, X_{i,n})}{\text{Var}_i(X_{i,n})} = \kappa_i^{\text{avg}} \frac{\text{Cov}_i(\bar{P}_{i,n}, X_{i,n})}{\text{Var}_i(X_{i,n})} + \frac{\text{Cov}_i(e_{i,n}^f, X_{i,n})}{\text{Var}_i(X_{i,n})}.$$

By the definitions in (26) the left side is κ_i^{eff} and the first ratio on the right is r_i^{int} , which is (28). The identity uses no orthogonality condition. It holds for any within-contract distribution with $\text{Var}_i(X_{i,n}) > 0$, and the product form obtains exactly when $\text{Cov}_i(e_{i,n}^f, X_{i,n}) = 0$. If the implemented active-region derivative equals the nominal derivative, then $\kappa_i^{\text{avg}} = \kappa_i^{\text{tpl}}$.

Without local linearity or orthogonality, write $f_{i,n}^{8,\text{fwd}} = G_i(\bar{P}_{i,n}) + \omega_{i,n}^f$. The population coefficient is

$$\frac{\text{Cov}_i(G_i(\bar{P}_{i,n}), X_{i,n}) + \text{Cov}_i(\omega_{i,n}^f, X_{i,n})}{\text{Var}_i(X_{i,n})}.$$

There is then no reason for the residual component to be orthogonal to $X_{i,n}$ and no universal factorization follows. $\square$